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**Mitsumori et al.**

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(54) **IMAGING APPARATUS, IMAGING APPARATUS CONTROL METHOD, AND PROGRAM**

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(72) Inventors: **Ryo Mitsumori**, Tokyo (JP); **Jiro Kawano**, Tokyo (JP); **Mizuho Hara**, Tokyo (JP); **Ryosuke Takeuchi**, Tokyo (JP)

(73) Assignee: **SONY GROUP CORPORATION**, Tokyo (JP)

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§ 371 (c)(1),
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PCT Pub. Date: **Sep. 29, 2022**

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(30) **Foreign Application Priority Data**
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(51) **Int. Cl.**
***H04N 23/69*** (2023.01)
***G06F 3/04845*** (2022.01)
(Continued)

(52) **U.S. Cl.**
CPC ......... ***H04N 23/69*** (2023.01); ***G06F 3/04845*** (2013.01); ***G06F 3/04847*** (2013.01); ***G06T 3/40*** (2013.01)

(58) **Field of Classification Search**
CPC ...... H04N 23/69; H04N 23/45; H04N 23/631; H04N 23/633; H04N 23/634;
(Continued)

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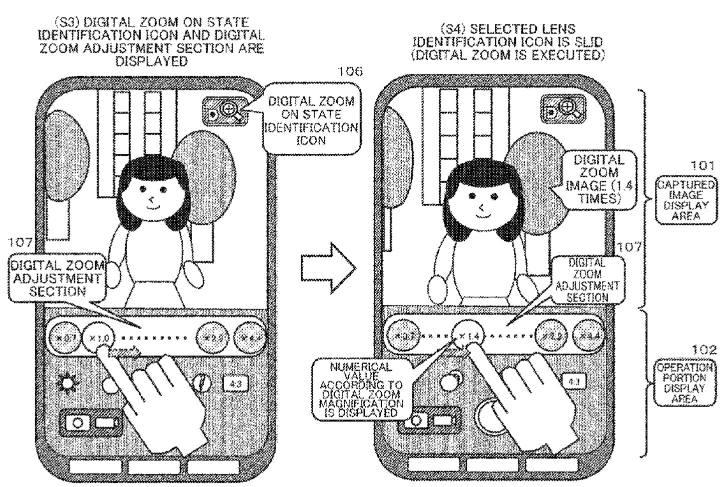
(74) *Attorney, Agent, or Firm* — CHIP LAW GROUP

(57) **ABSTRACT**

An operation portion (UI) that enables efficient execution of photographing lens selection processing by a user and digital zoom processing for a captured image using a selected lens is realized. A plurality of lenses with different lens magnifications available for image capturing, and a display control unit configured to execute control of display data to be output to a display unit are included, and the display control unit displays a plurality of lens selection portions corresponding to the plurality of respective lenses on the display unit, displays a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable
(Continued)



depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit, and displays a digital zoom image executed depending on the slide amount on the display unit.

**14 Claims, 34 Drawing Sheets**

(51) **Int. Cl.**
***G06F 3/04847*** (2022.01)
***G06T 3/40*** (2024.01)

(58) **Field of Classification Search**
CPC ............. G06F 3/04845; G06F 3/04847; G06F 1/1686; G06F 3/0482; G06F 3/04817; G06F 3/04883; G06T 3/40; G03B 19/22; G03B 30/00
See application file for complete search history.

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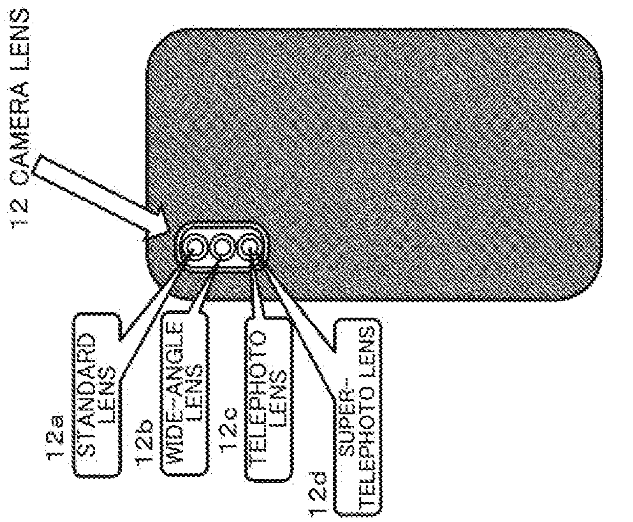

12 CAMERA LENS
12a STANDARD LENS
12b WIDE-ANGLE LENS
12c TELEPHOTO LENS
12d SUPER-TELEPHOTO LENS
BACK SURFACE

Fig. 1B

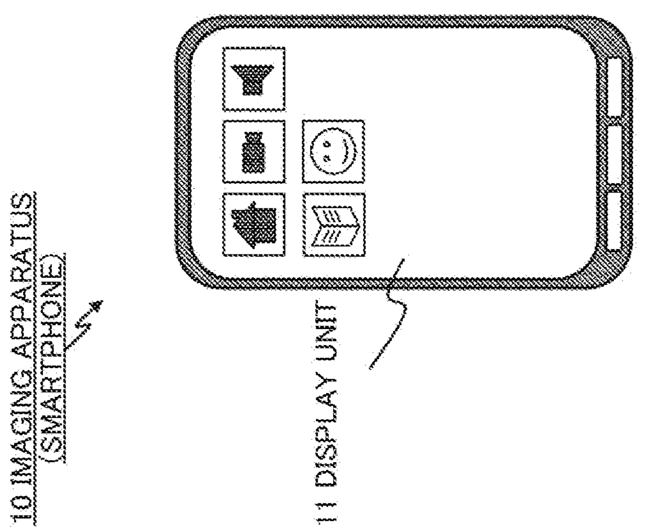

10 IMAGING APPARATUS (SMARTPHONE)
11 DISPLAY UNIT
FRONT SURFACE

Fig. 1A

Fig. 2A

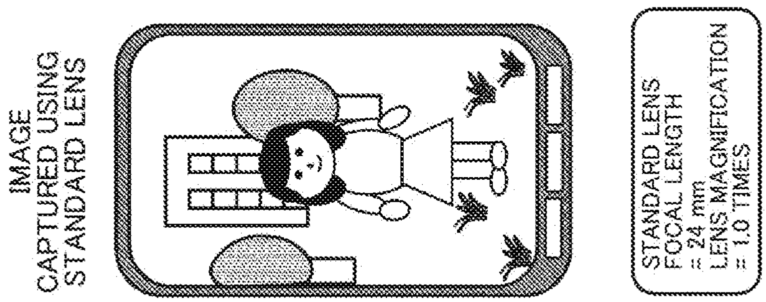



Fig. 2B

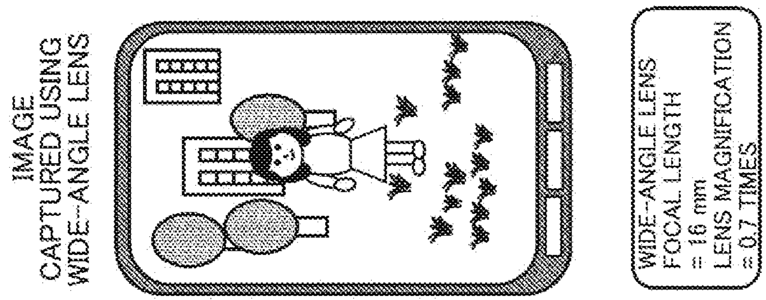



Fig. 2C

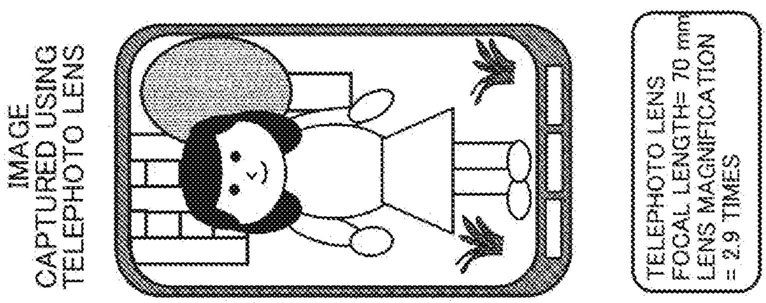
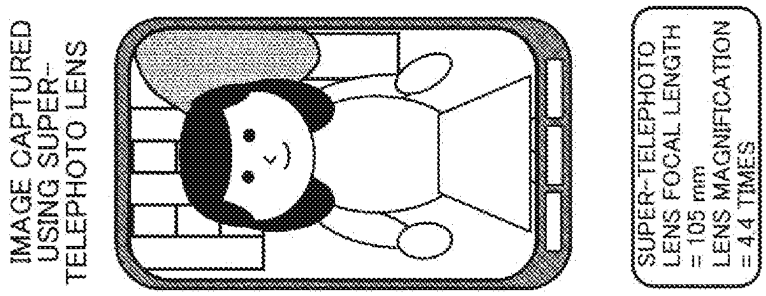



Fig. 2D



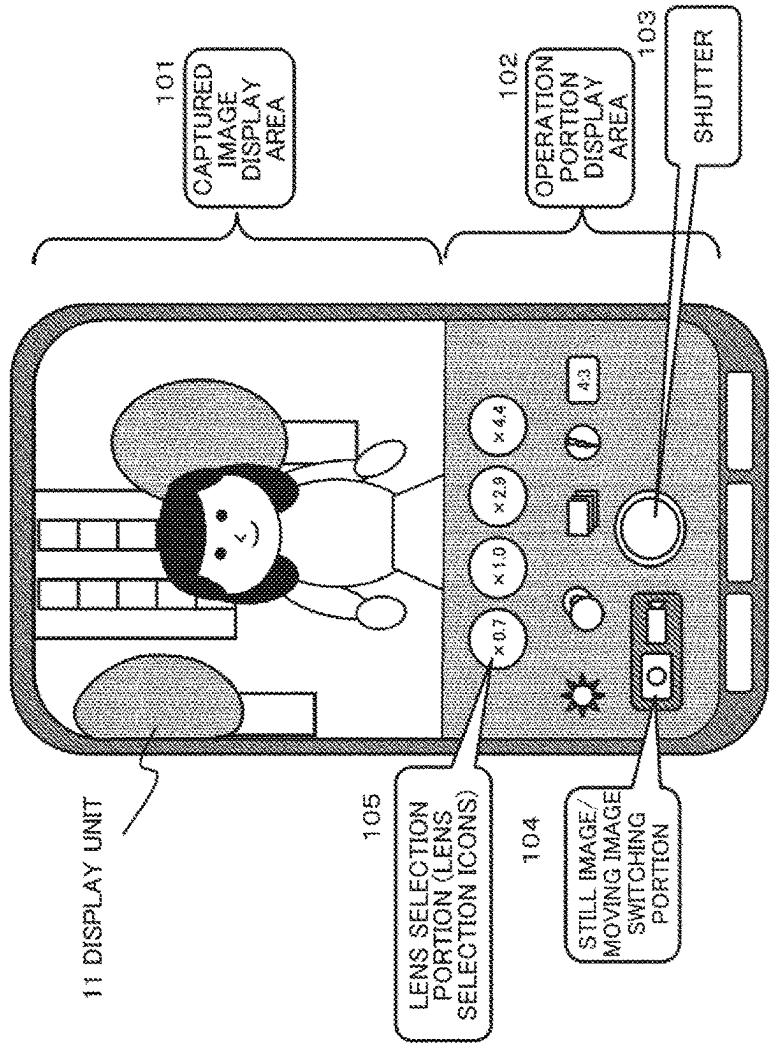

101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA
103
SHUTTER
11 DISPLAY UNIT
105
LENS SELECTION PORTION (LENS SELECTION ICONS)
104
STILL IMAGE/ MOVING IMAGE SWITCHING PORTION
×0.7
×1.0
×2.9
×4.4
4:3

Fig. 3

Fig. 4A

IMAGE CAPTURED USING STANDARD LENS

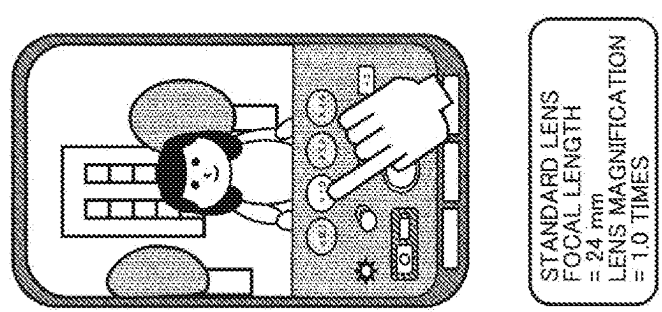



Fig. 4B

IMAGE CAPTURED USING WIDE-ANGLE LENS

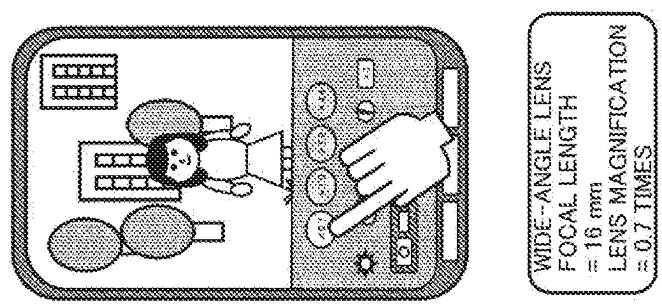



Fig. 4C

IMAGE CAPTURED USING TELEPHOTO LENS

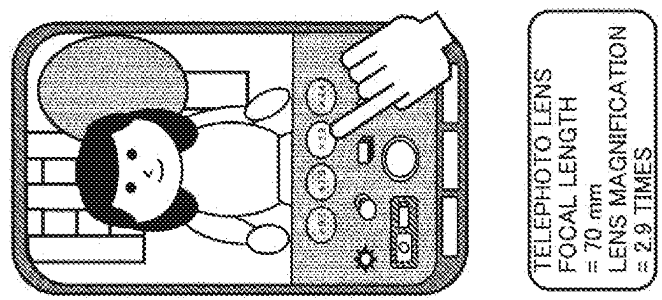
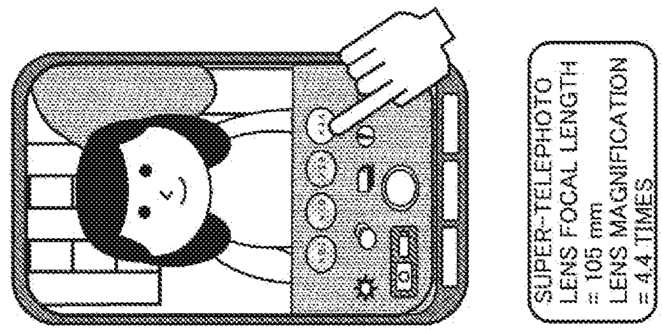



Fig. 4D

IMAGE CAPTURED USING SUPER-TELEPHOTO LENS



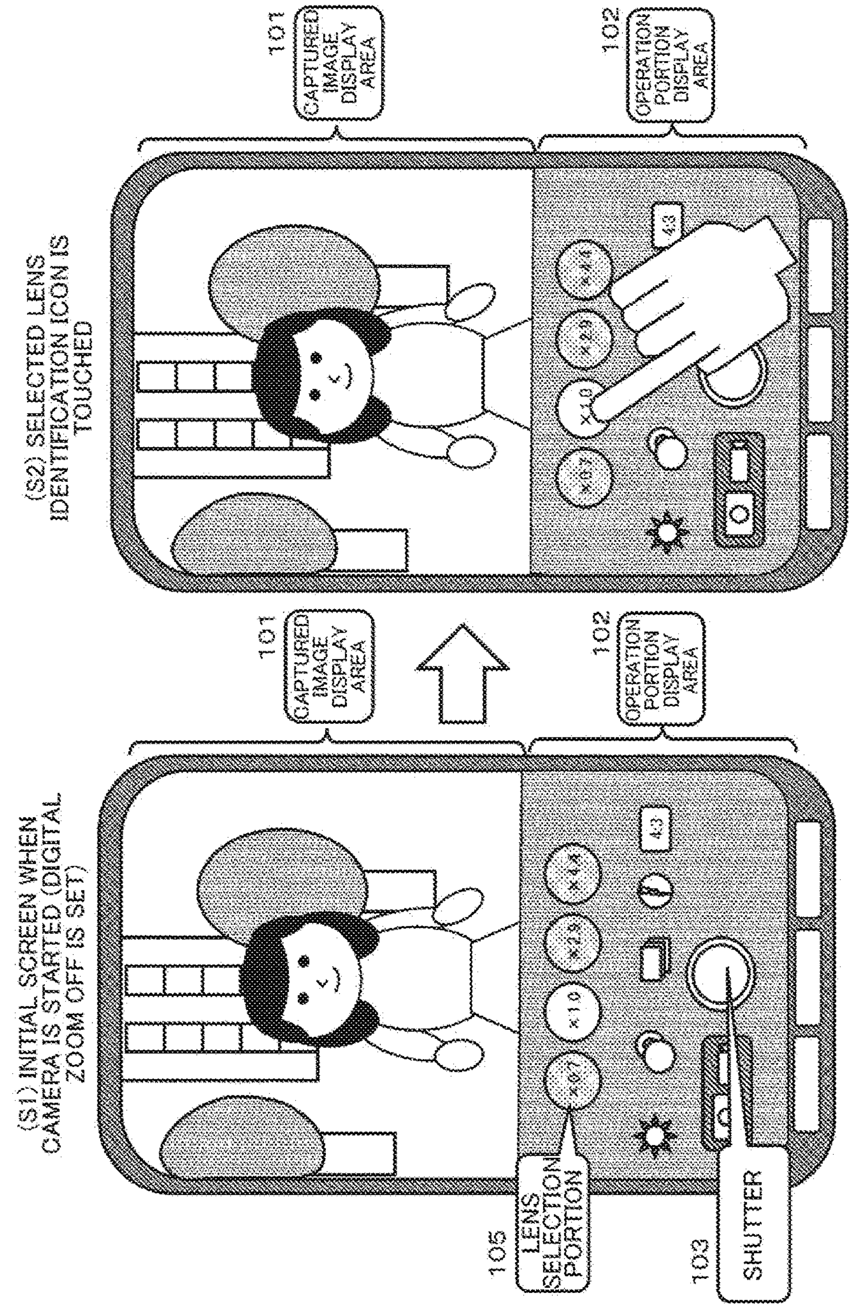

(S1) INITIAL SCREEN WHEN CAMERA IS STARTED (DIGITAL ZOOM OFF IS SET)
(S2) SELECTED LENS IDENTIFICATION ICON IS TOUCHED
101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA
101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA
105
LENS SELECTION PORTION
103
SHUTTER
×0.7
×1.0
×2.9
×4.4
4:3

Fig. 5

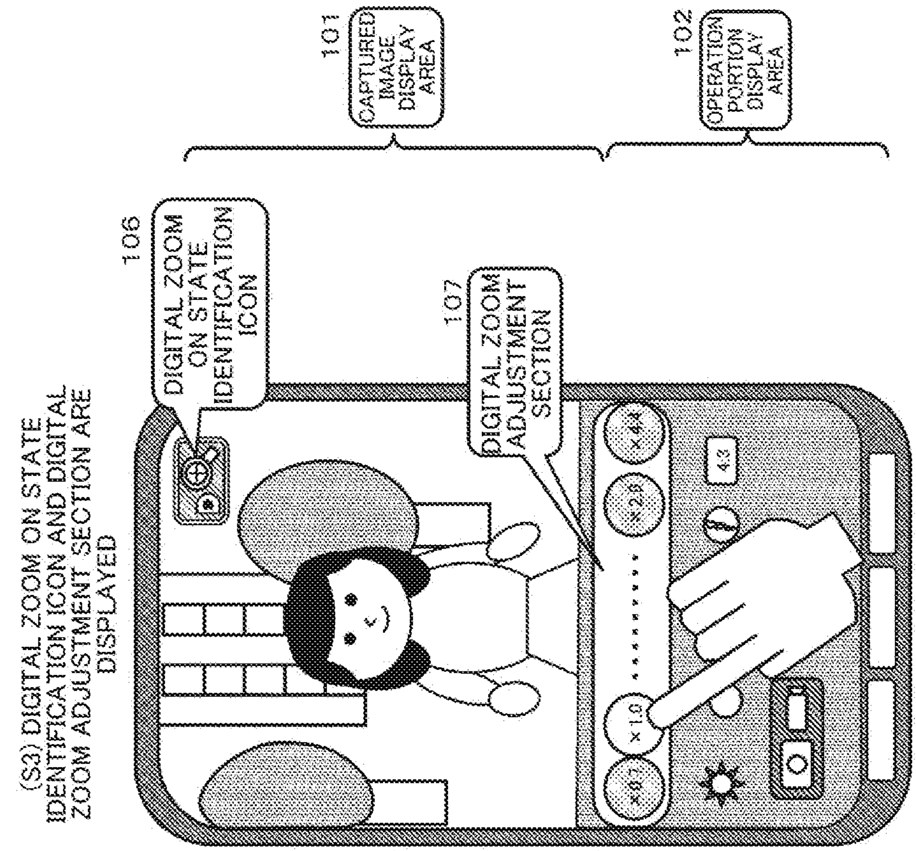

(S3) DIGITAL ZOOM ON STATE IDENTIFICATION ICON AND DIGITAL ZOOM ADJUSTMENT SECTION ARE DISPLAYED
106
DIGITAL ZOOM ON STATE IDENTIFICATION ICON
107
DIGITAL ZOOM ADJUSTMENT SECTION
101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA
×1.0
×2.6
×4.4
4:3

Fig. 6

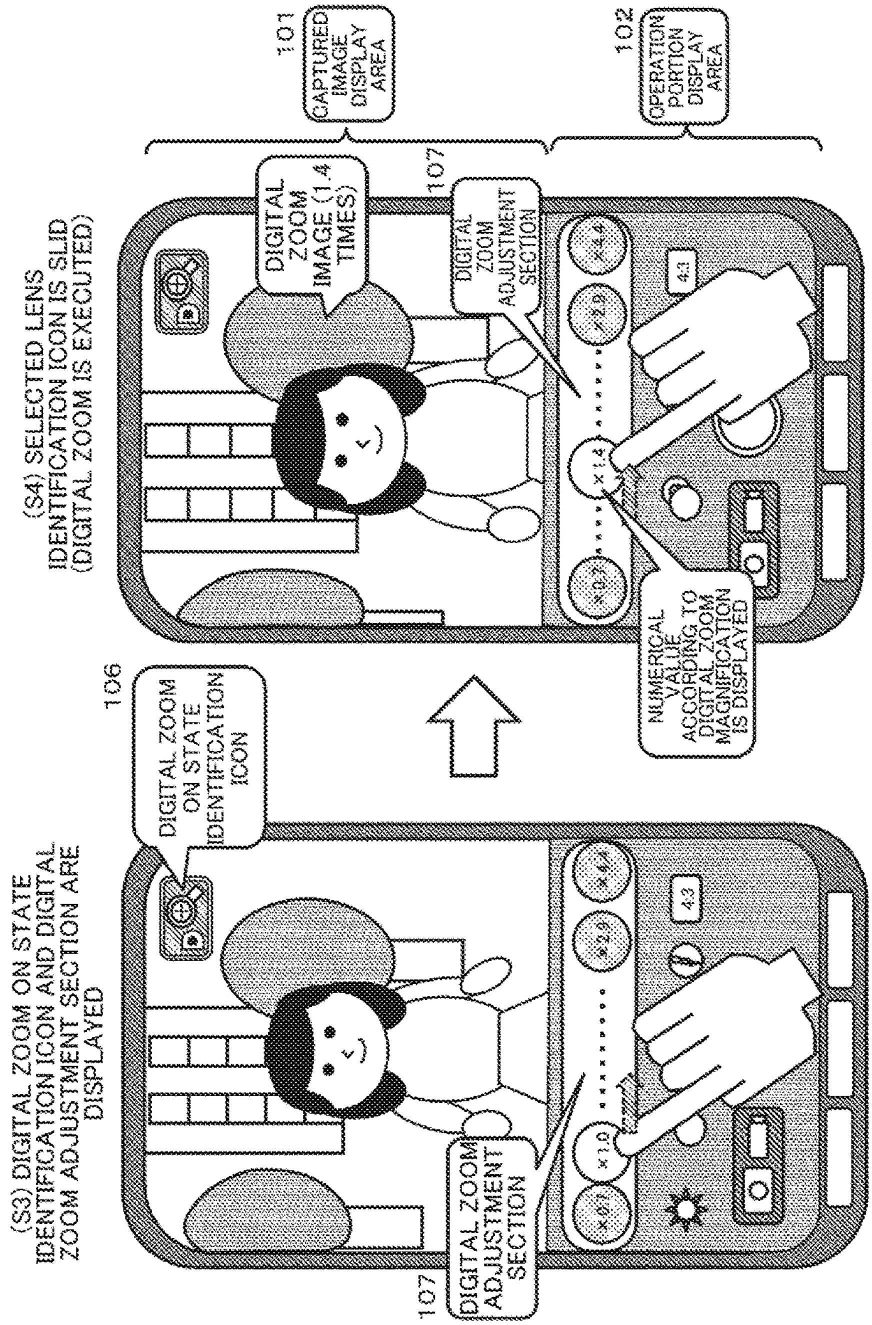

(S3) DIGITAL ZOOM ON STATE IDENTIFICATION ICON AND DIGITAL ZOOM ADJUSTMENT SECTION ARE DISPLAYED
106
DIGITAL ZOOM ON STATE IDENTIFICATION ICON
107
DIGITAL ZOOM ADJUSTMENT SECTION
(S4) SELECTED LENS IDENTIFICATION ICON IS SLID (DIGITAL ZOOM IS EXECUTED)
DIGITAL ZOOM IMAGE (1.4 TIMES)
107
DIGITAL ZOOM ADJUSTMENT SECTION
NUMERICAL VALUE ACCORDING TO DIGITAL ZOOM MAGNIFICATION IS DISPLAYED
101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA

Fig. 7

(S5) SELECTED LENS IDENTIFICATION ICON IS SLID (DIGITAL ZOOM IS EXECUTED)

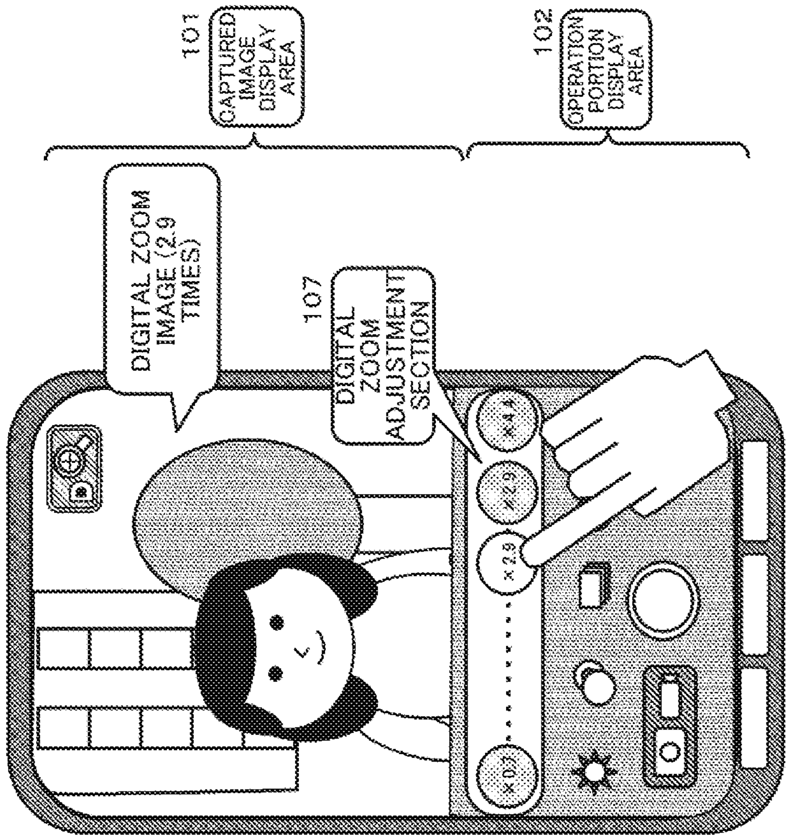



Fig. 8

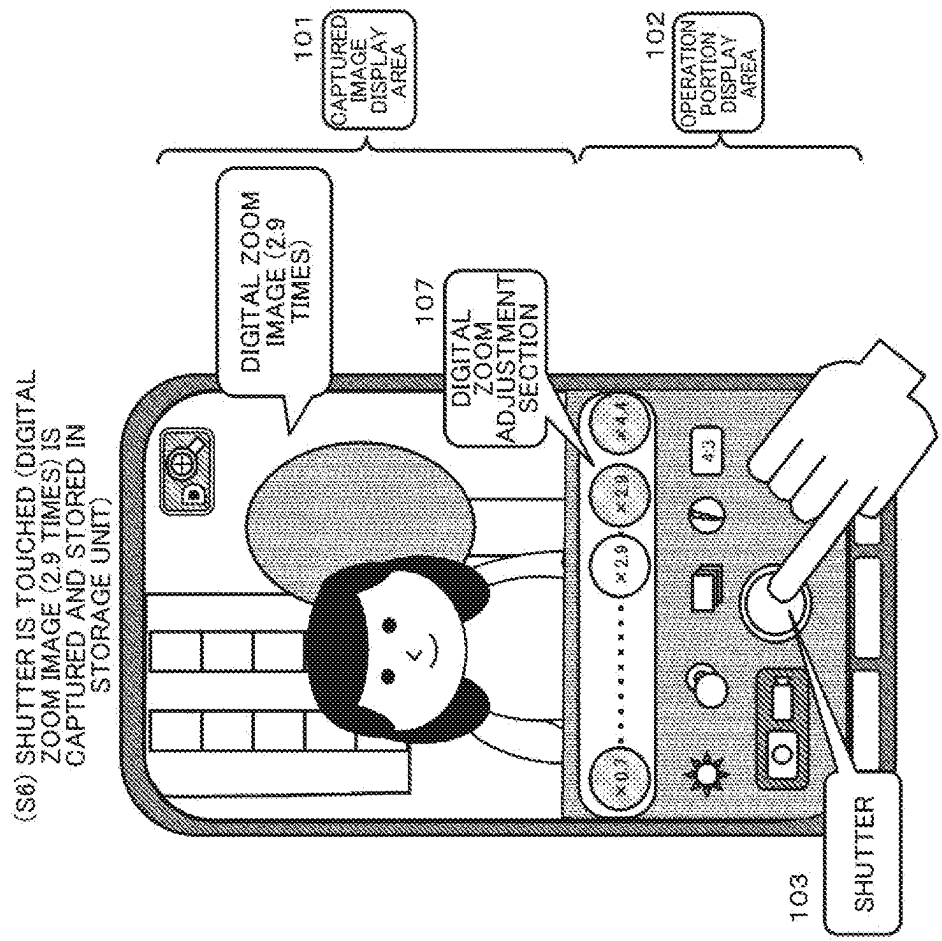

(S6) SHUTTER IS TOUCHED (DIGITAL ZOOM IMAGE (2.9 TIMES) IS CAPTURED AND STORED IN STORAGE UNIT)
DIGITAL ZOOM IMAGE (2.9 TIMES)
107
DIGITAL ZOOM ADJUSTMENT SECTION
101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA
×2.9
103
SHUTTER

Fig. 9

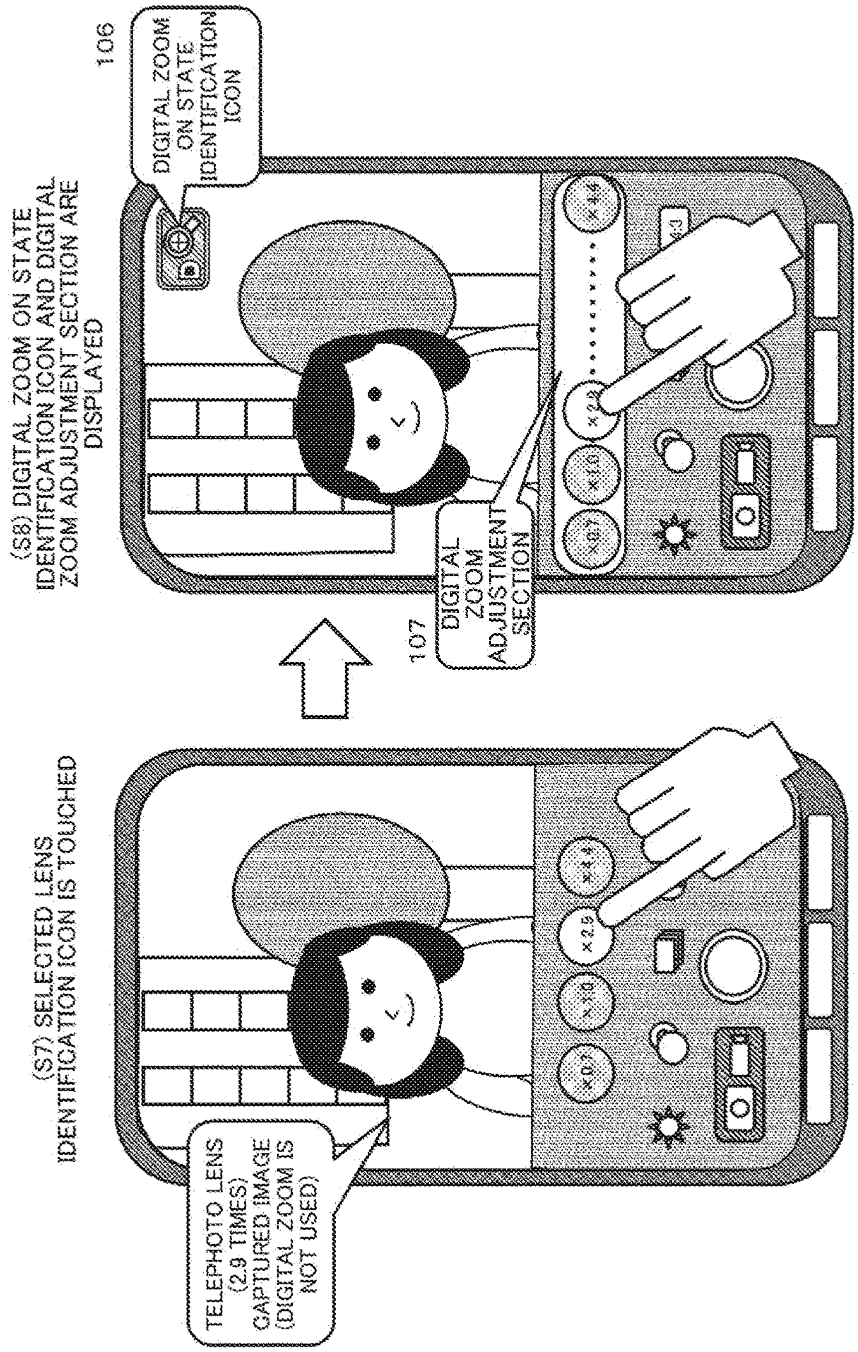

(S7) SELECTED LENS IDENTIFICATION ICON IS TOUCHED
TELEPHOTO LENS (2.9 TIMES) CAPTURED IMAGE (DIGITAL ZOOM IS NOT USED)
(S8) DIGITAL ZOOM ON STATE IDENTIFICATION ICON AND DIGITAL ZOOM ADJUSTMENT SECTION ARE DISPLAYED
106
DIGITAL ZOOM ON STATE IDENTIFICATION ICON
107
DIGITAL ZOOM ADJUSTMENT SECTION

Fig. 10

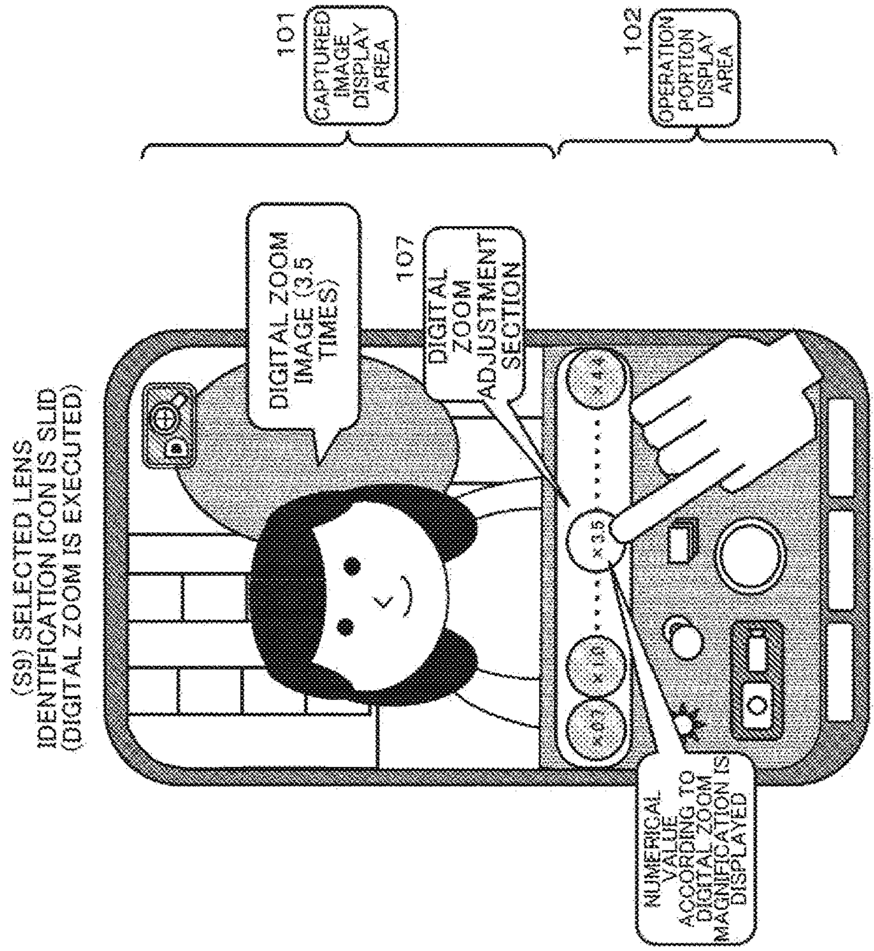

(S9) SELECTED LENS
IDENTIFICATION ICON IS SLID
(DIGITAL ZOOM IS EXECUTED)
101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA
DIGITAL ZOOM IMAGE (3.5 TIMES)
107
DIGITAL ZOOM ADJUSTMENT SECTION
×3.5
NUMERICAL VALUE ACCORDING TO DIGITAL ZOOM MAGNIFICATION IS DISPLAYED

Fig. 11

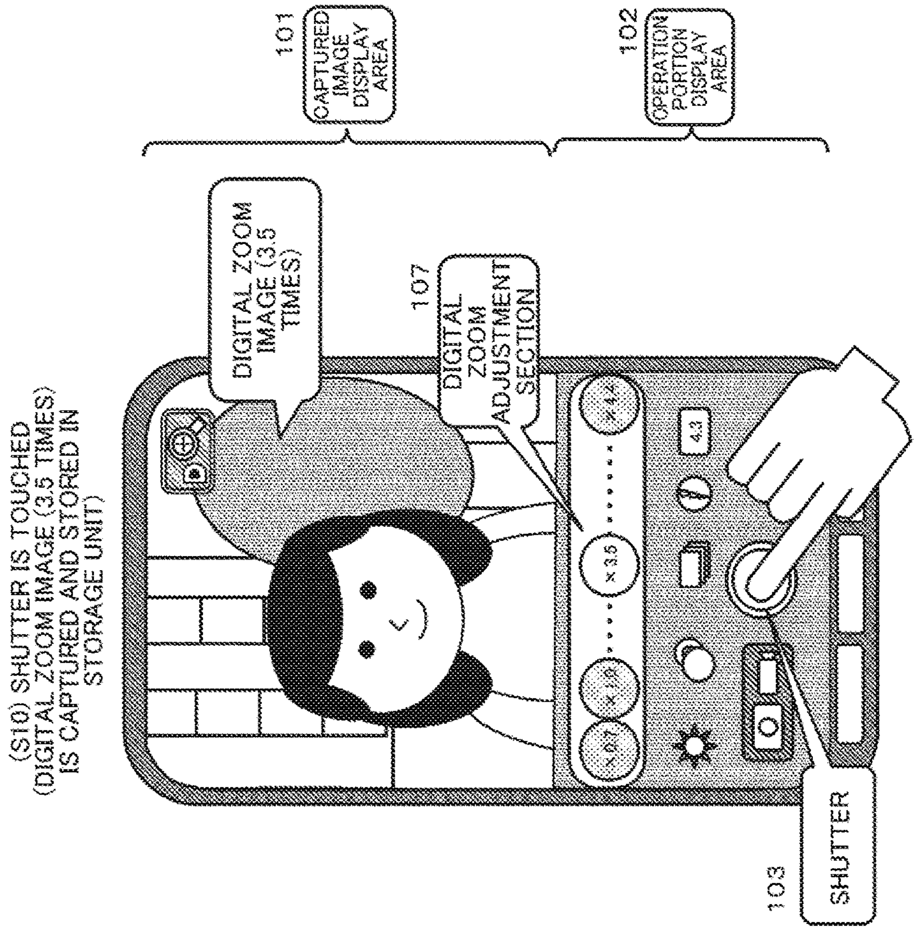
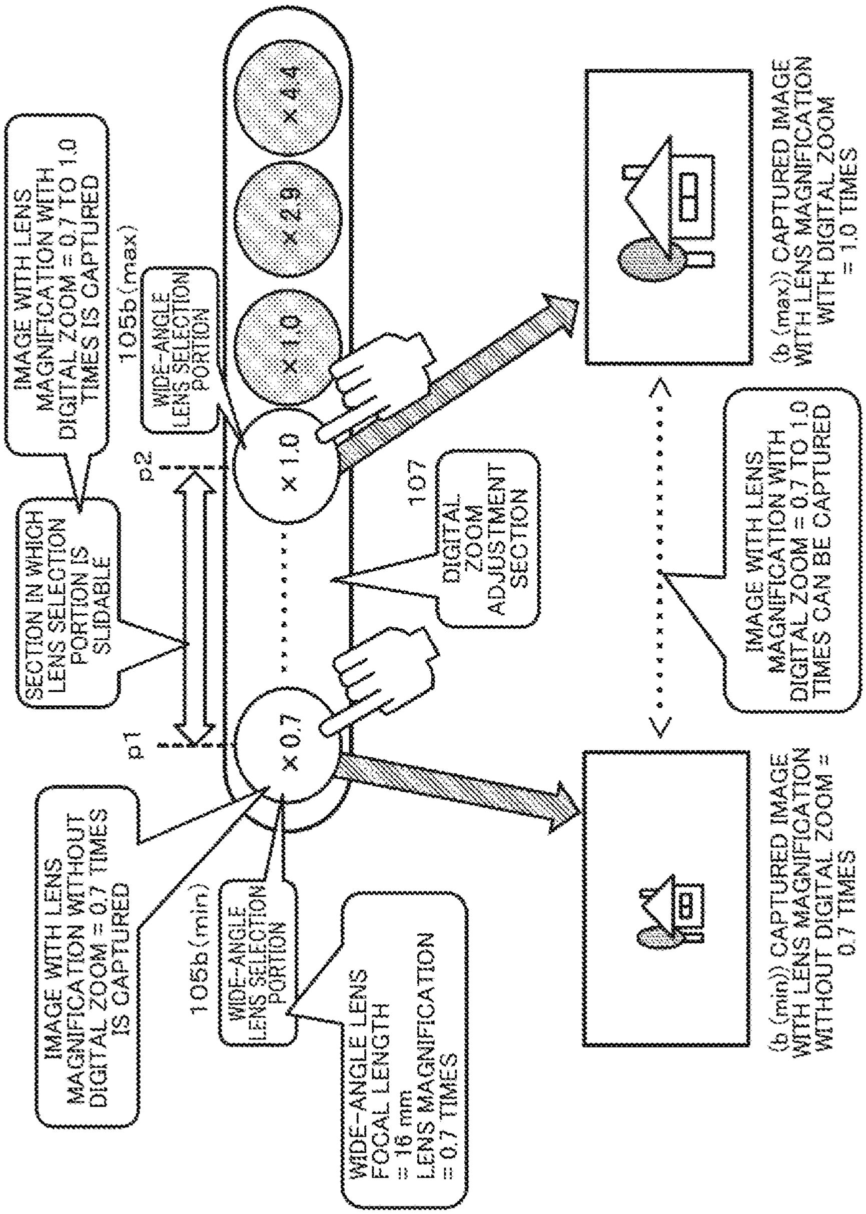

(S10) SHUTTER IS TOUCHED
(DIGITAL ZOOM IMAGE (3.5 TIMES)
IS CAPTURED AND STORED IN
STORAGE UNIT)
101
CAPTURED IMAGE DISPLAY AREA
102
OPERATION PORTION DISPLAY AREA
DIGITAL ZOOM IMAGE (3.5 TIMES)
107
DIGITAL ZOOM ADJUSTMENT SECTION
×3.5
103
SHUTTER

Fig. 12



105b(max)
IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 0.7 TO 1.0 TIMES IS CAPTURED
WIDE-ANGLE LENS SELECTION PORTION
×1.0
×2.9
×4.4
p2
×1.0
SECTION IN WHICH LENS SELECTION PORTION IS SLIDABLE
107
DIGITAL ZOOM ADJUSTMENT SECTION
p1
×0.7
IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 0.7 TO 1.0 TIMES CAN BE CAPTURED
(b (max)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 1.0 TIMES
IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 0.7 TIMES IS CAPTURED
105b(min)
WIDE-ANGLE LENS SELECTION PORTION
WIDE-ANGLE LENS FOCAL LENGTH = 16 mm LENS MAGNIFICATION = 0.7 TIMES
(b (min)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 0.7 TIMES


Fig. 13

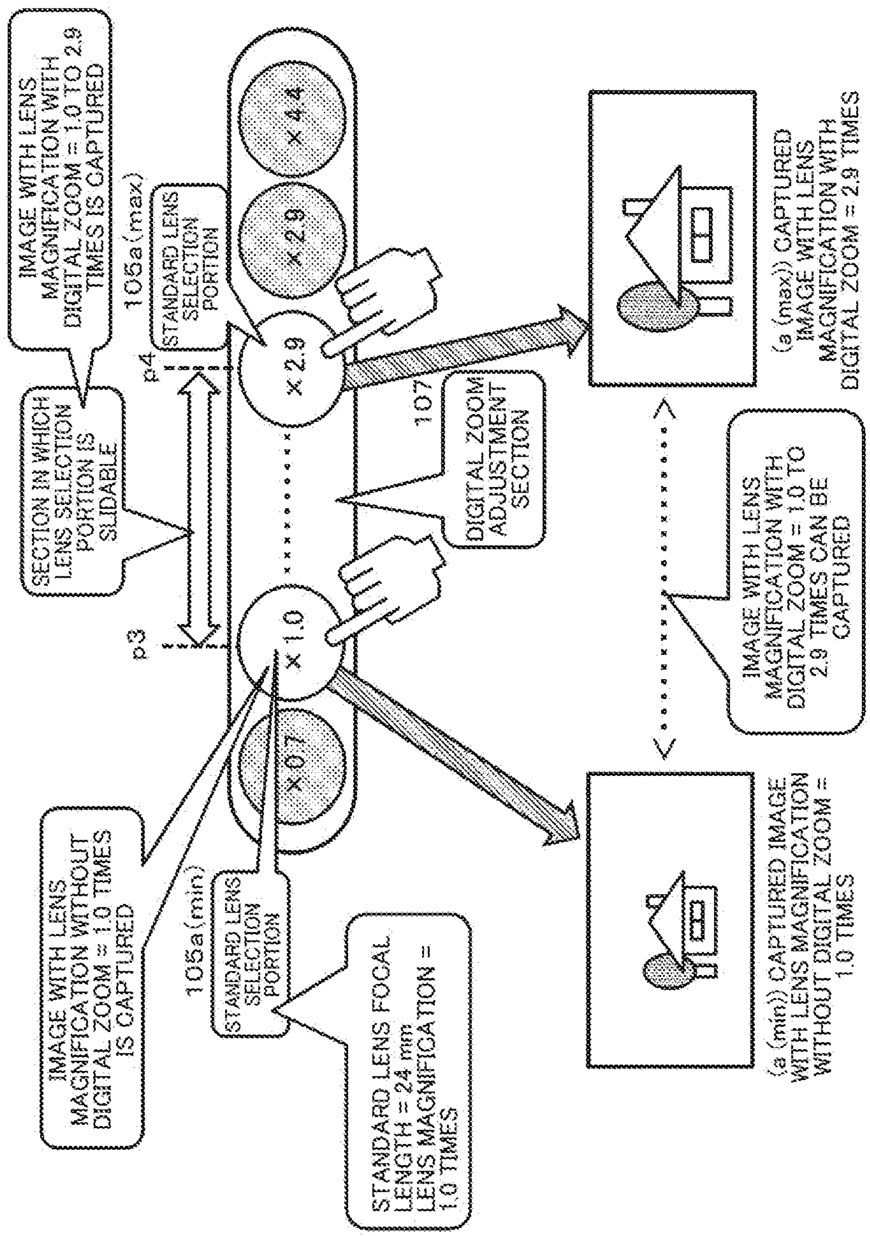


IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 1.0 TO 2.9 TIMES IS CAPTURED
105a (max)
STANDARD LENS SELECTION PORTION
p4
p3
SECTION IN WHICH LENS SELECTION PORTION IS SLIDABLE
×4.4
×2.9
×2.9
×1.0
×0.7
107
DIGITAL ZOOM ADJUSTMENT SECTION
(a (max)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 2.9 TIMES
IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 1.0 TO 2.9 TIMES CAN BE CAPTURED
(a (min)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 1.0 TIMES
IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 1.0 TIMES IS CAPTURED
105a (min)
STANDARD LENS SELECTION PORTION
STANDARD LENS FOCAL LENGTH = 24 mm LENS MAGNIFICATION = 1.0 TIMES

Fig. 14

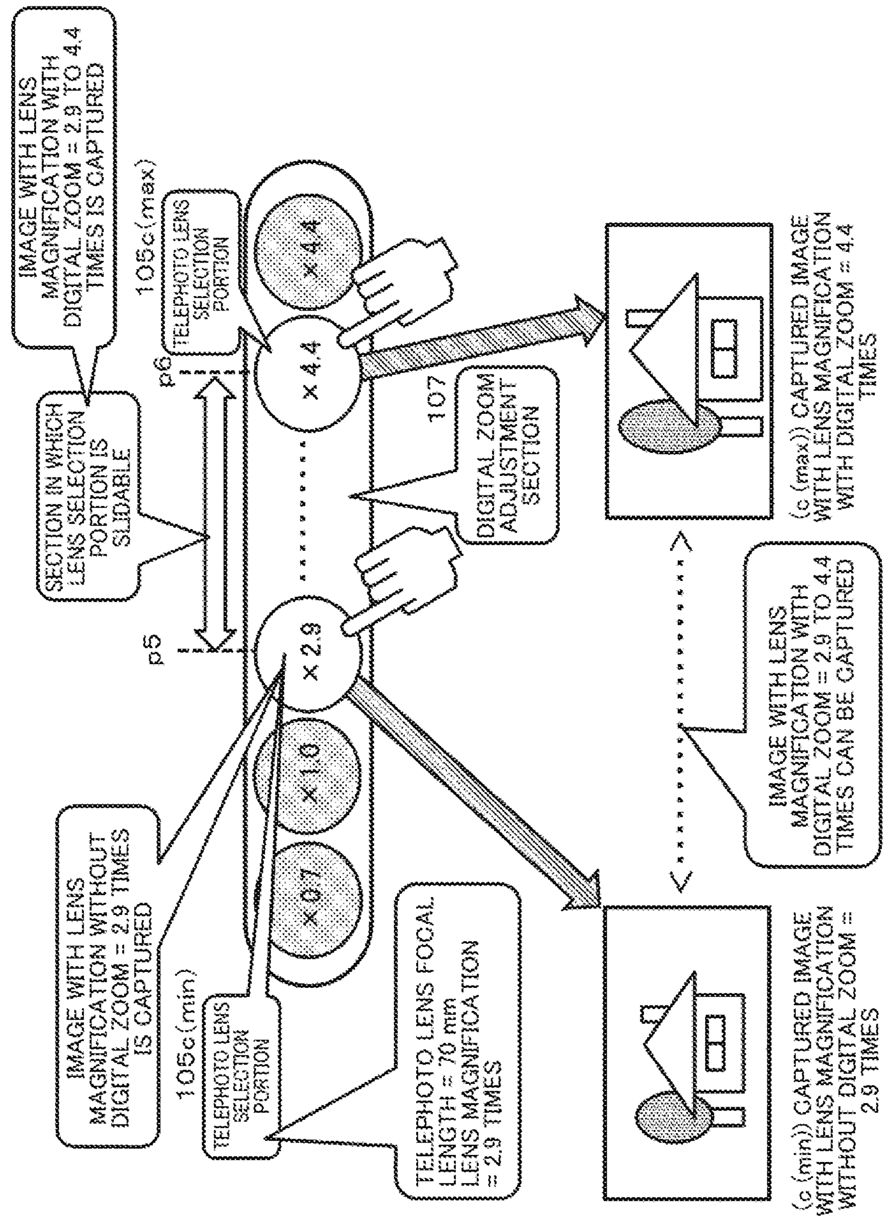


IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 2.9 TO 4.4 TIMES IS CAPTURED
SECTION IN WHICH LENS SELECTION PORTION IS SLIDABLE
105c(max)
TELEPHOTO LENS SELECTION PORTION
p6
p5
×4.4
×4.4
×2.9
×1.0
×0.7
107
DIGITAL ZOOM ADJUSTMENT SECTION
IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 2.9 TIMES IS CAPTURED
105c(min)
TELEPHOTO LENS SELECTION PORTION
TELEPHOTO LENS FOCAL LENGTH = 70 mm LENS MAGNIFICATION = 2.9 TIMES
IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 2.9 TO 4.4 TIMES CAN BE CAPTURED
(c(max)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 4.4 TIMES
(c(min)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 2.9 TIMES

Fig. 15

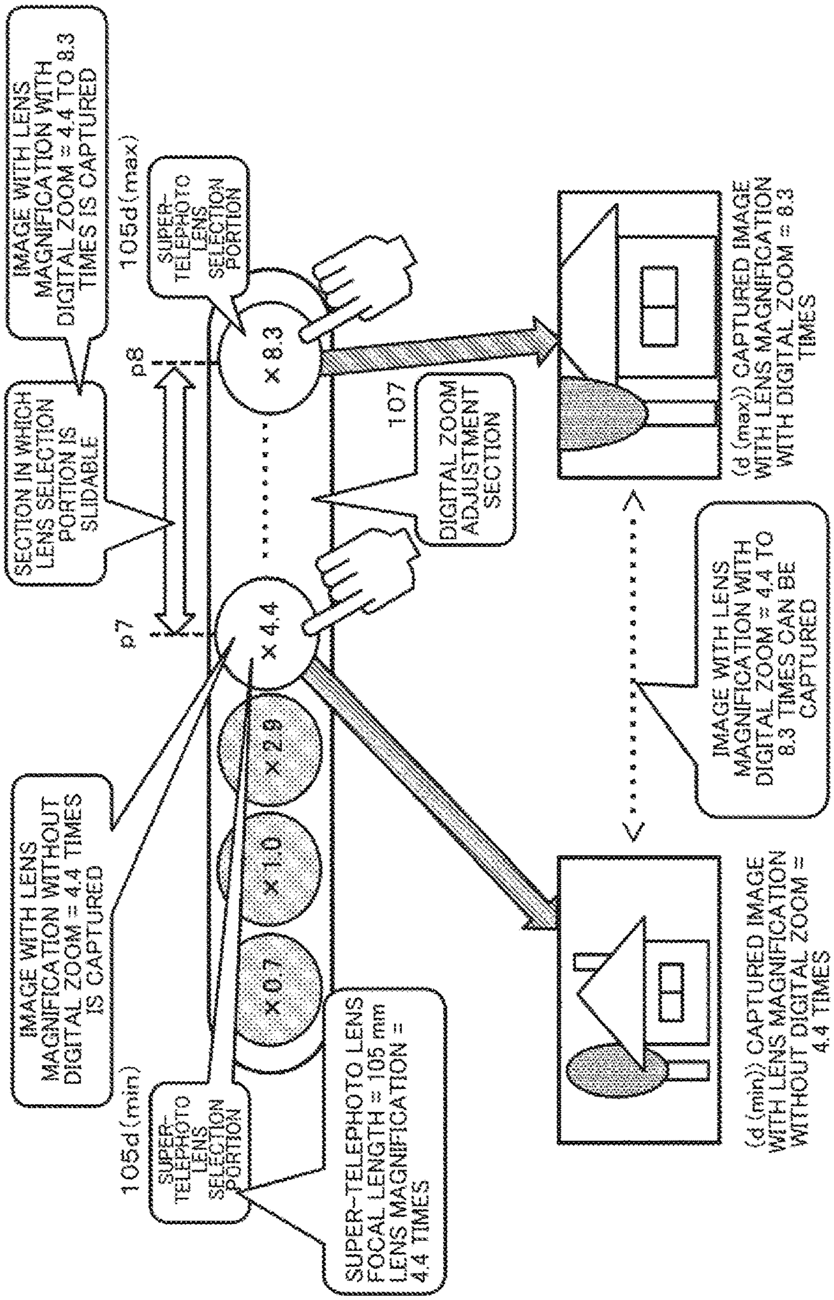


IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 4.4 TO 8.3 TIMES IS CAPTURED
105d(max)
SUPER-TELEPHOTO LENS SELECTION PORTION
SECTION IN WHICH LENS SELECTION PORTION IS SLIDABLE
p8
p7
×8.3
×4.4
×2.9
×1.0
×0.7
107
DIGITAL ZOOM ADJUSTMENT SECTION
(d (max)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 8.3 TIMES
IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 4.4 TO 8.3 TIMES CAN BE CAPTURED
IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 4.4 TIMES IS CAPTURED
105d(min)
SUPER-TELEPHOTO LENS SELECTION PORTION
SUPER-TELEPHOTO LENS FOCAL LENGTH = 105 mm LENS MAGNIFICATION = 4.4 TIMES
(d (min)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 4.4 TIMES

Fig. 16

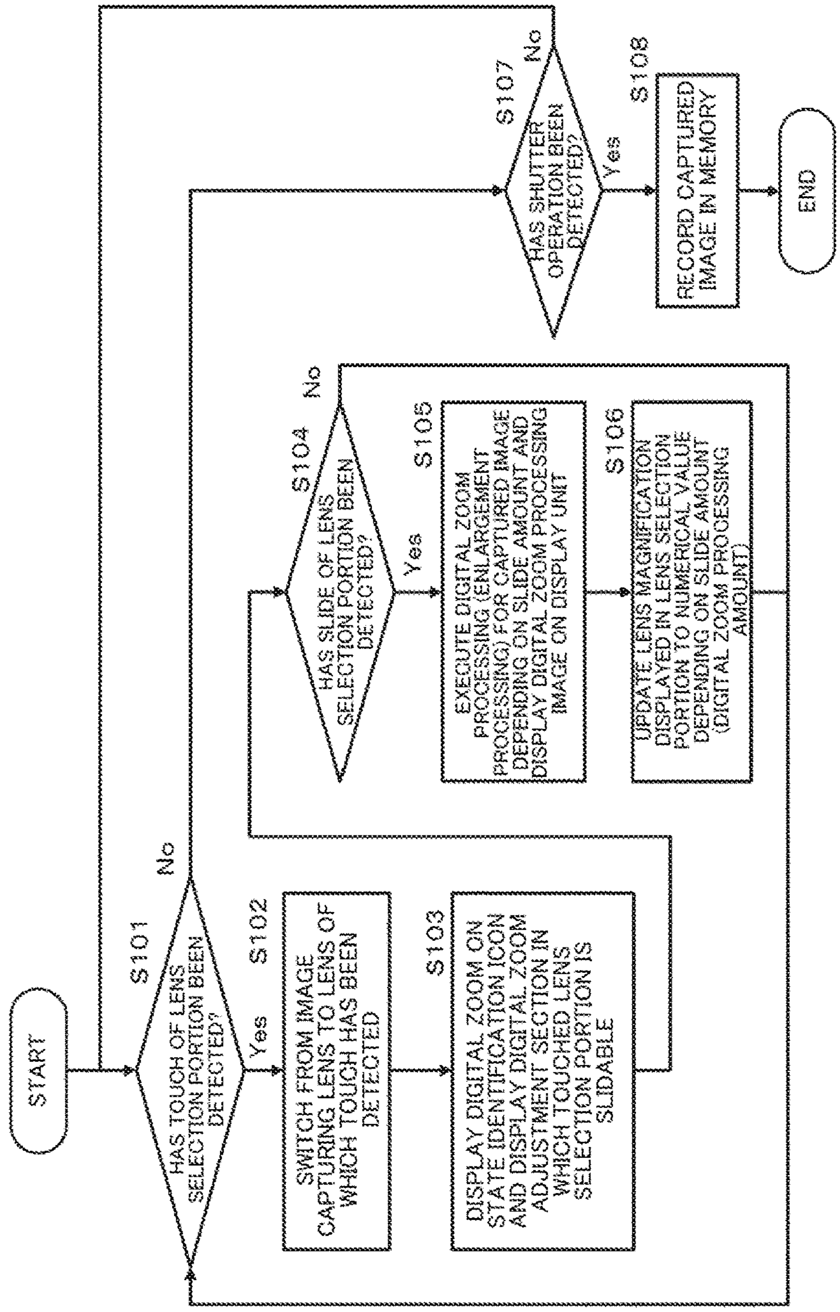

START
S101
HAS TOUCH OF LENS SELECTION PORTION BEEN DETECTED?
Yes
No
S102
SWITCH FROM IMAGE CAPTURING LENS TO LENS OF WHICH TOUCH HAS BEEN DETECTED
S103
DISPLAY DIGITAL ZOOM ON STATE IDENTIFICATION ICON AND DISPLAY DIGITAL ZOOM ADJUSTMENT SECTION IN WHICH TOUCHED LENS SELECTION PORTION IS SLIDABLE
S104
HAS SLIDE OF LENS SELECTION PORTION BEEN DETECTED?
Yes
No
S105
EXECUTE DIGITAL ZOOM PROCESSING (ENLARGEMENT PROCESSING) FOR CAPTURED IMAGE DEPENDING ON SLIDE AMOUNT AND DISPLAY DIGITAL ZOOM PROCESSING IMAGE ON DISPLAY UNIT
S106
UPDATE LENS MAGNIFICATION DISPLAYED IN LENS SELECTION PORTION TO NUMERICAL VALUE DEPENDING ON SLIDE AMOUNT (DIGITAL ZOOM PROCESSING AMOUNT)
S107
HAS SHUTTER OPERATION BEEN DETECTED?
Yes
No
S108
RECORD CAPTURED IMAGE IN MEMORY
END

Fig. 17

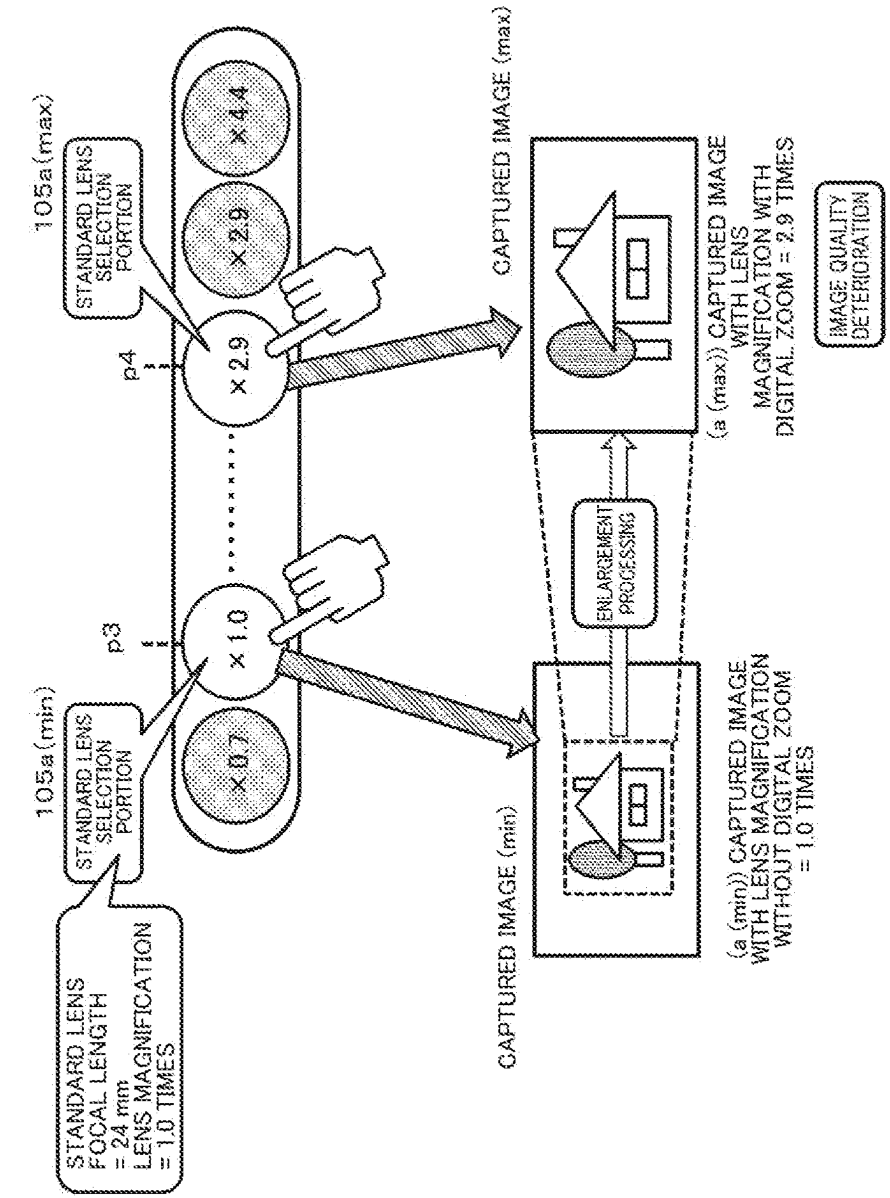
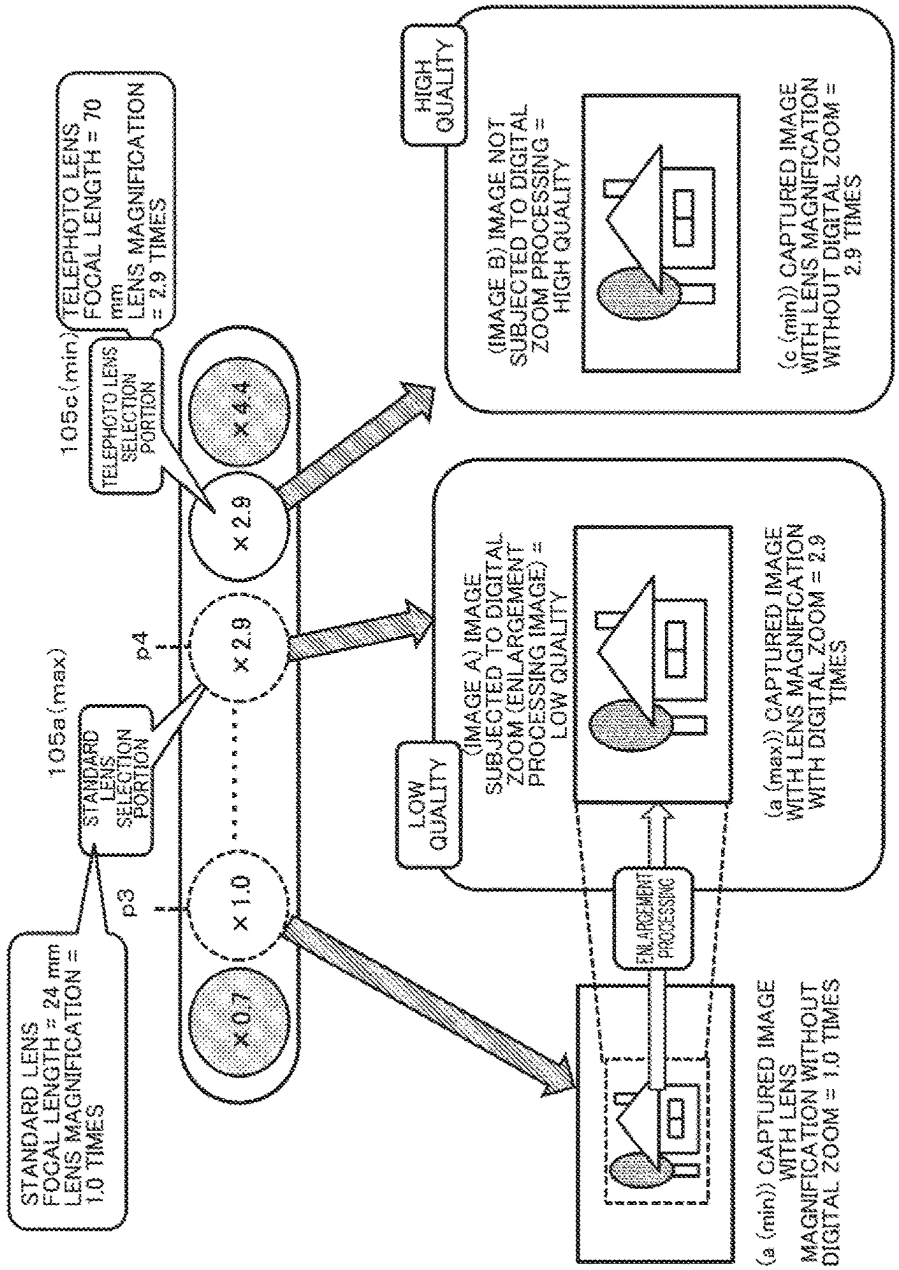

STANDARD LENS FOCAL LENGTH = 24 mm LENS MAGNIFICATION = 1.0 TIMES
105a (min)
STANDARD LENS SELECTION PORTION
p3
×0.7
×1.0
p4
×2.9
105a (max)
STANDARD LENS SELECTION PORTION
×2.9
×4.4
CAPTURED IMAGE (min)
(a (min)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 1.0 TIMES
ENLARGEMENT PROCESSING
CAPTURED IMAGE (max)
(a (max)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 2.9 TIMES
IMAGE QUALITY DETERIORATION

Fig. 18

Fig. 19


STANDARD LENS
FOCAL LENGTH = 24 mm
LENS MAGNIFICATION =
1.0 TIMES
p3
105a (max)
STANDARD LENS SELECTION PORTION
p4
105c (min)
TELEPHOTO LENS SELECTION PORTION
TELEPHOTO LENS
FOCAL LENGTH = 70 mm
LENS MAGNIFICATION
= 2.9 TIMES
× 0.7
× 1.0
× 2.9
× 2.9
× 4.4
(a (min)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 1.0 TIMES
ENLARGEMENT PROCESSING
LOW QUALITY
(IMAGE A) IMAGE SUBJECTED TO DIGITAL ZOOM (ENLARGEMENT PROCESSING IMAGE) = LOW QUALITY
(a (max)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITH DIGITAL ZOOM = 2.9 TIMES
HIGH QUALITY
(IMAGE B) IMAGE NOT SUBJECTED TO DIGITAL ZOOM PROCESSING = HIGH QUALITY
(c (min)) CAPTURED IMAGE WITH LENS MAGNIFICATION WITHOUT DIGITAL ZOOM = 2.9 TIMES

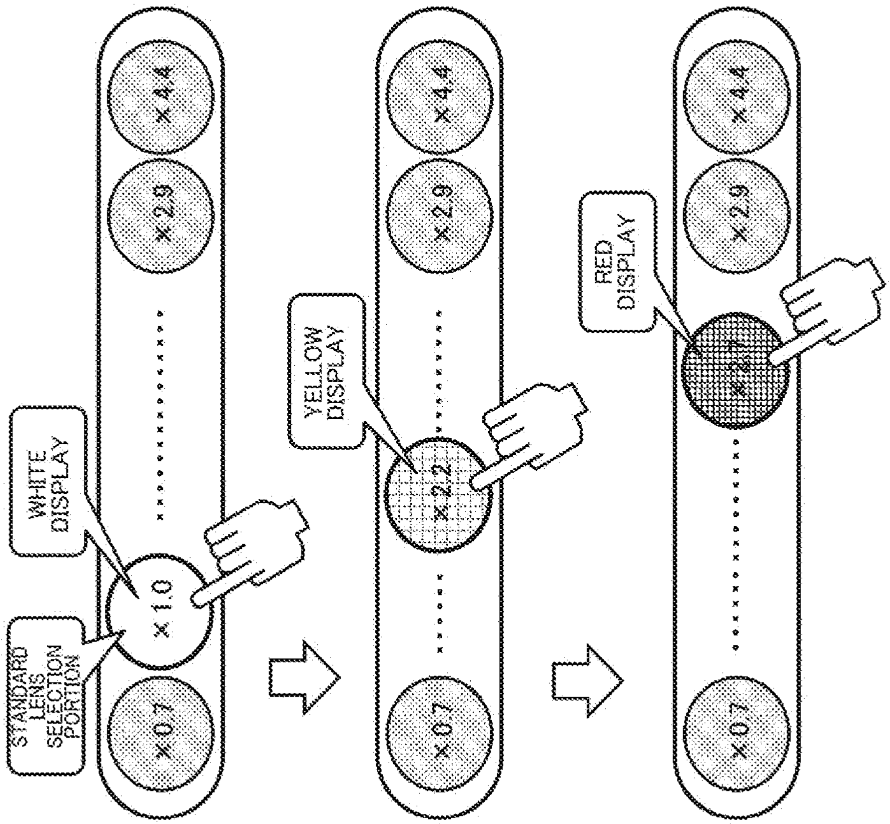



Fig. 20A NO DIGITAL ZOOM (NO IMAGE QUALITY DETERIORATION)

Fig. 20B AMOUNT OF DIGITAL ZOOM = SMALL (IMAGE QUALITY DETERIORATION = SMALL)

Fig. 20C AMOUNT OF DIGITAL ZOOM = LARGE (IMAGE QUALITY DETERIORATION = LARGE)

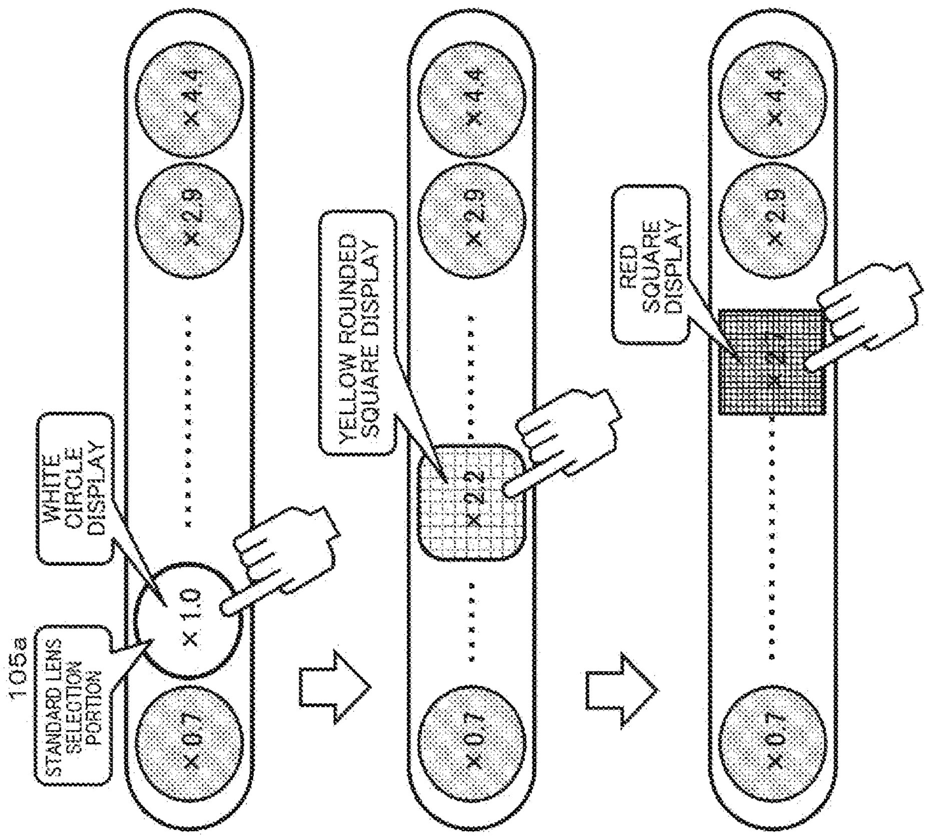



Fig. 21A NO DIGITAL ZOOM (NO IMAGE QUALITY DETERIORATION)

Fig. 21B AMOUNT OF DIGITAL ZOOM = SMALL (IMAGE QUALITY DETERIORATION = SMALL)

Fig. 21C AMOUNT OF DIGITAL ZOOM = LARGE (IMAGE QUALITY DETERIORATION = LARGE)

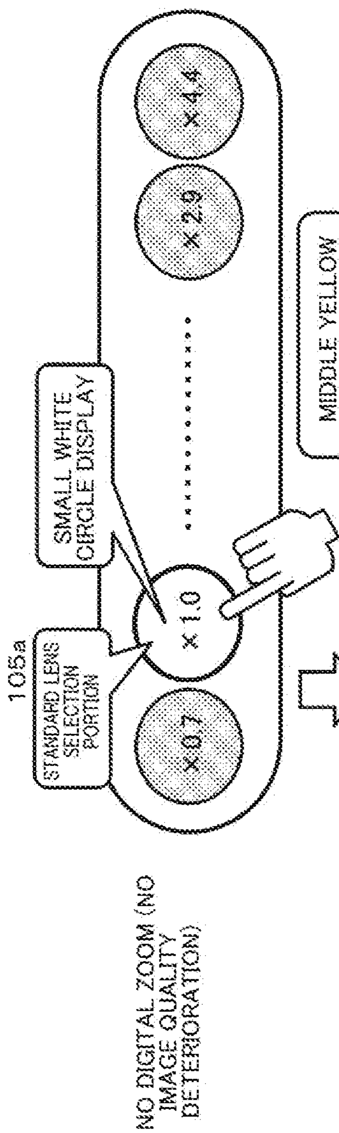



Fig. 22A

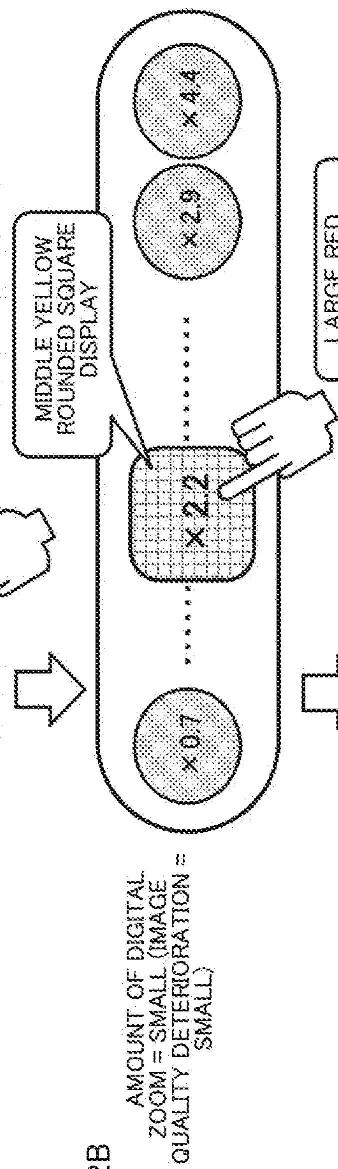



Fig. 22B

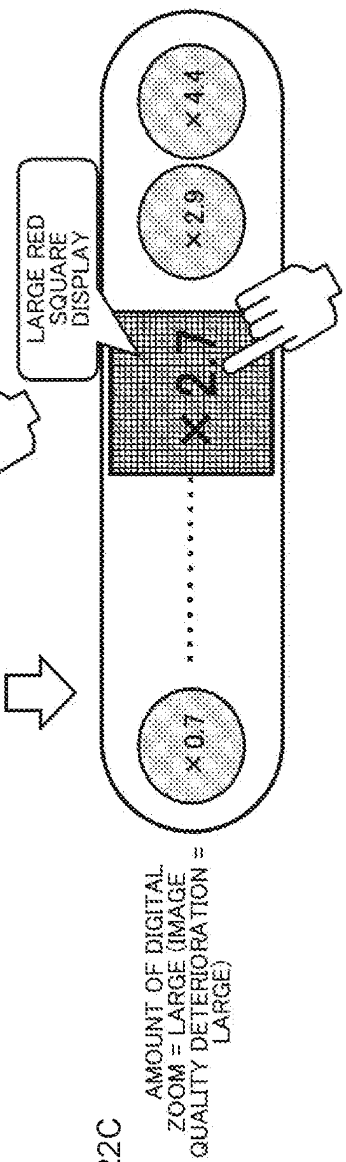



Fig. 22C

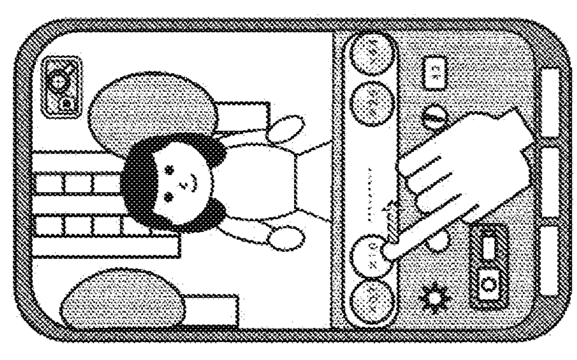


Fig. 23A

NO DIGITAL ZOOM (NO IMAGE QUALITY DETERIORATION)

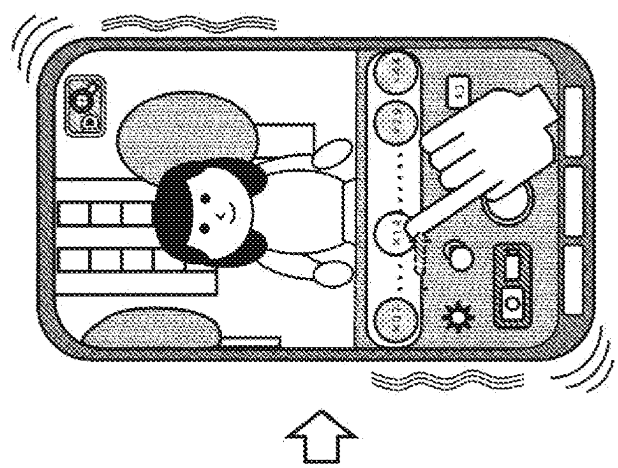


Fig. 23B

AMOUNT OF DIGITAL ZOOM = SMALL (IMAGE QUALITY DETERIORATION = SMALL)

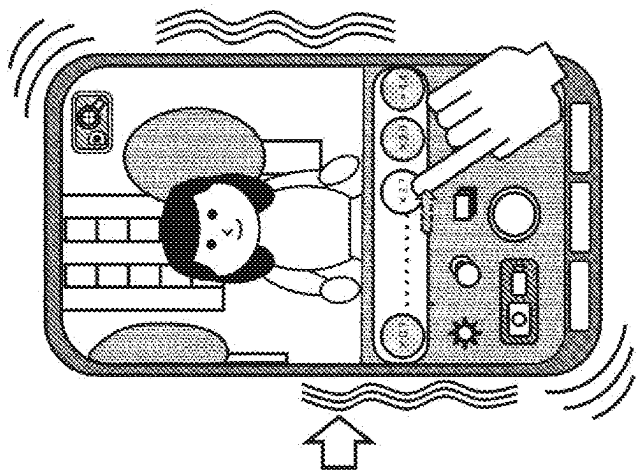


Fig. 23C

AMOUNT OF DIGITAL ZOOM = LARGE (IMAGE QUALITY DETERIORATION = LARGE)

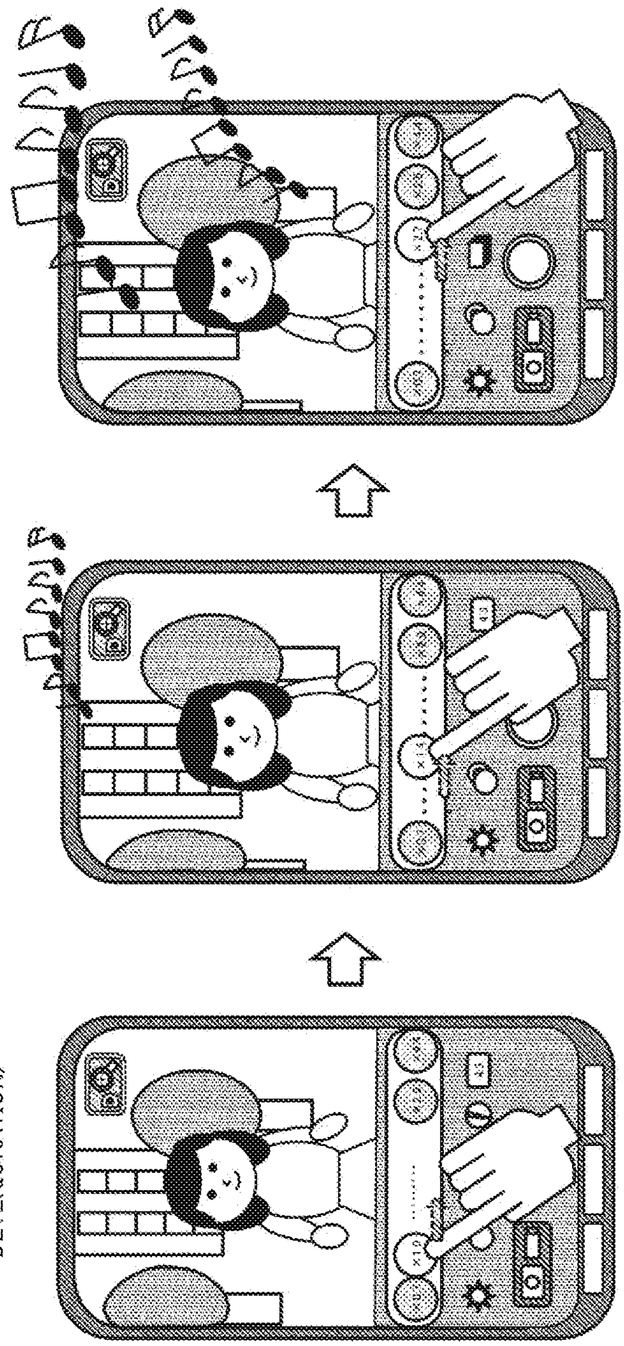


Fig. 24A
NO DIGITAL ZOOM (NO IMAGE QUALITY DETERIORATION)

Fig. 24B
AMOUNT OF DIGITAL ZOOM = SMALL (IMAGE QUALITY DETERIORATION = SMALL)

Fig. 24C
AMOUNT OF DIGITAL ZOOM = LARGE (IMAGE QUALITY DETERIORATION = LARGE)

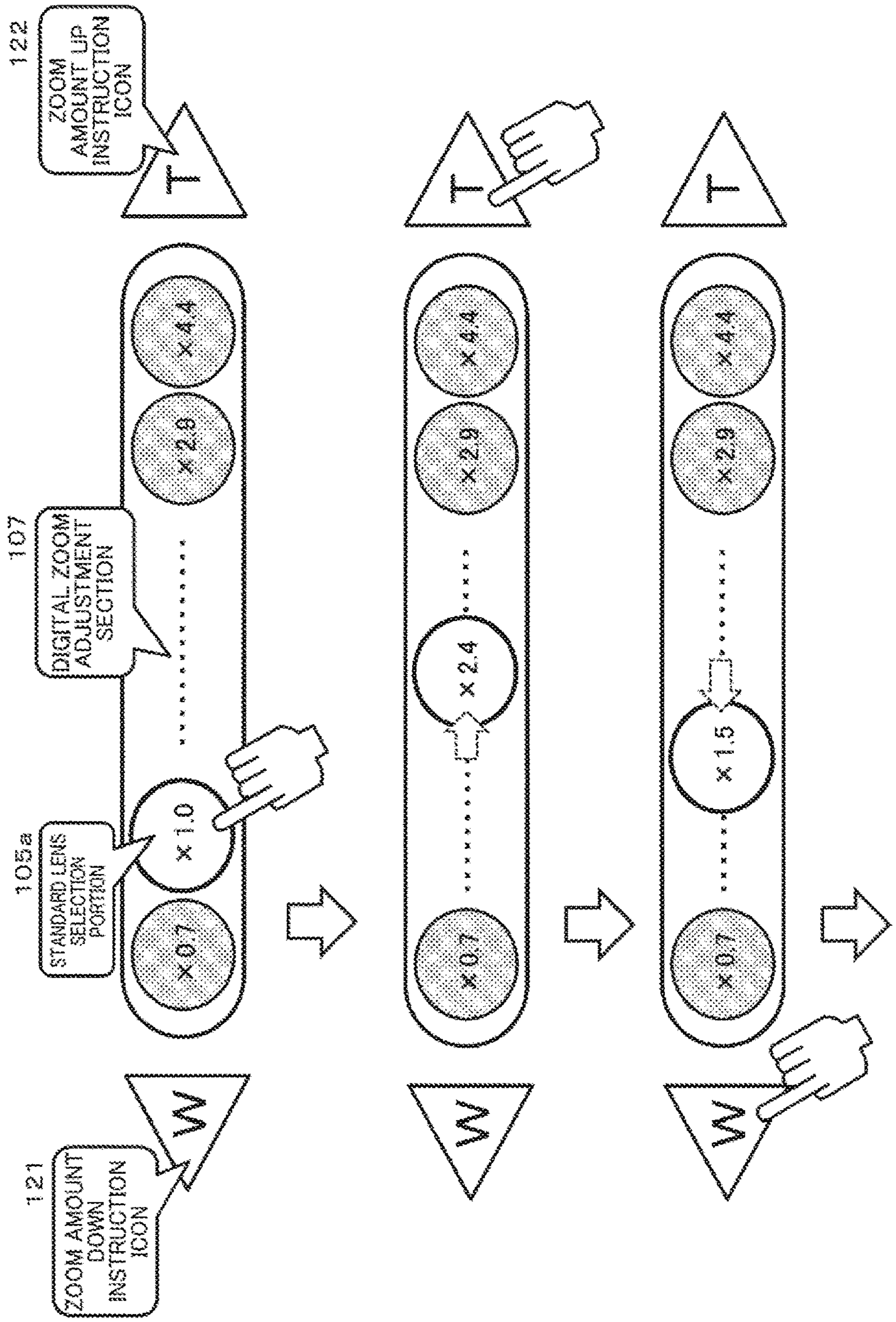

122
ZOOM AMOUNT UP INSTRUCTION ICON
T
×4.4
×2.9
107
DIGITAL ZOOM ADJUSTMENT SECTION
105a
STANDARD LENS SELECTION PORTION
×1.0
×0.7
W
121
ZOOM AMOUNT DOWN INSTRUCTION ICON
×2.4
×1.5

Fig. 25

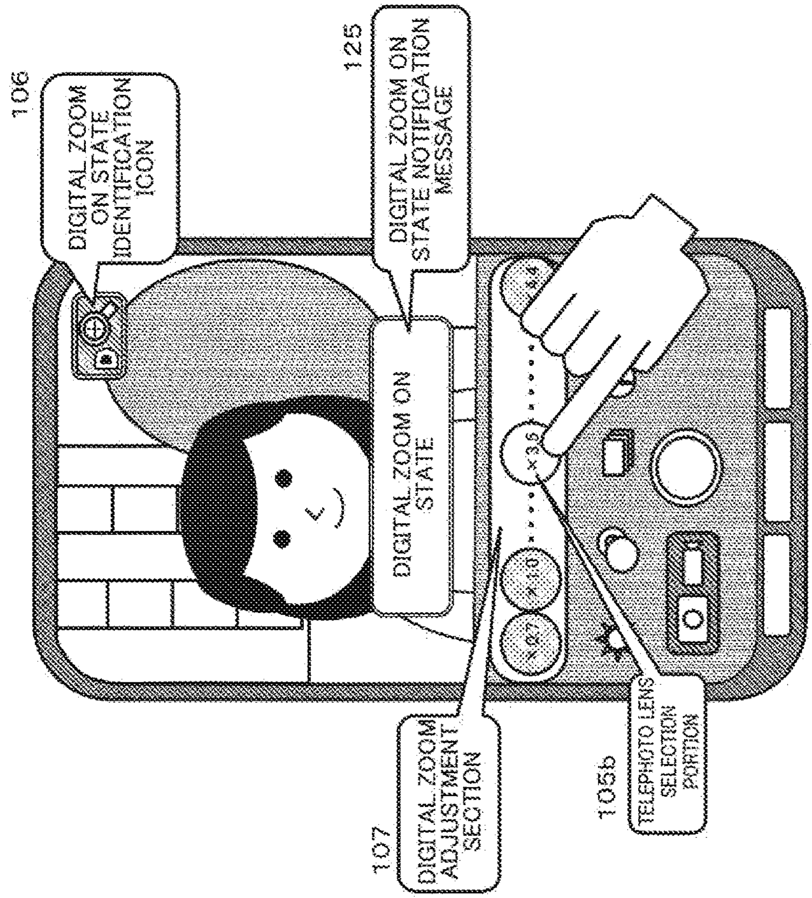

106
DIGITAL ZOOM ON STATE IDENTIFICATION ICON
125
DIGITAL ZOOM ON STATE NOTIFICATION MESSAGE
DIGITAL ZOOM ON STATE
107
DIGITAL ZOOM ADJUSTMENT SECTION
105b
TELEPHOTO LENS SELECTION PORTION

Fig. 26

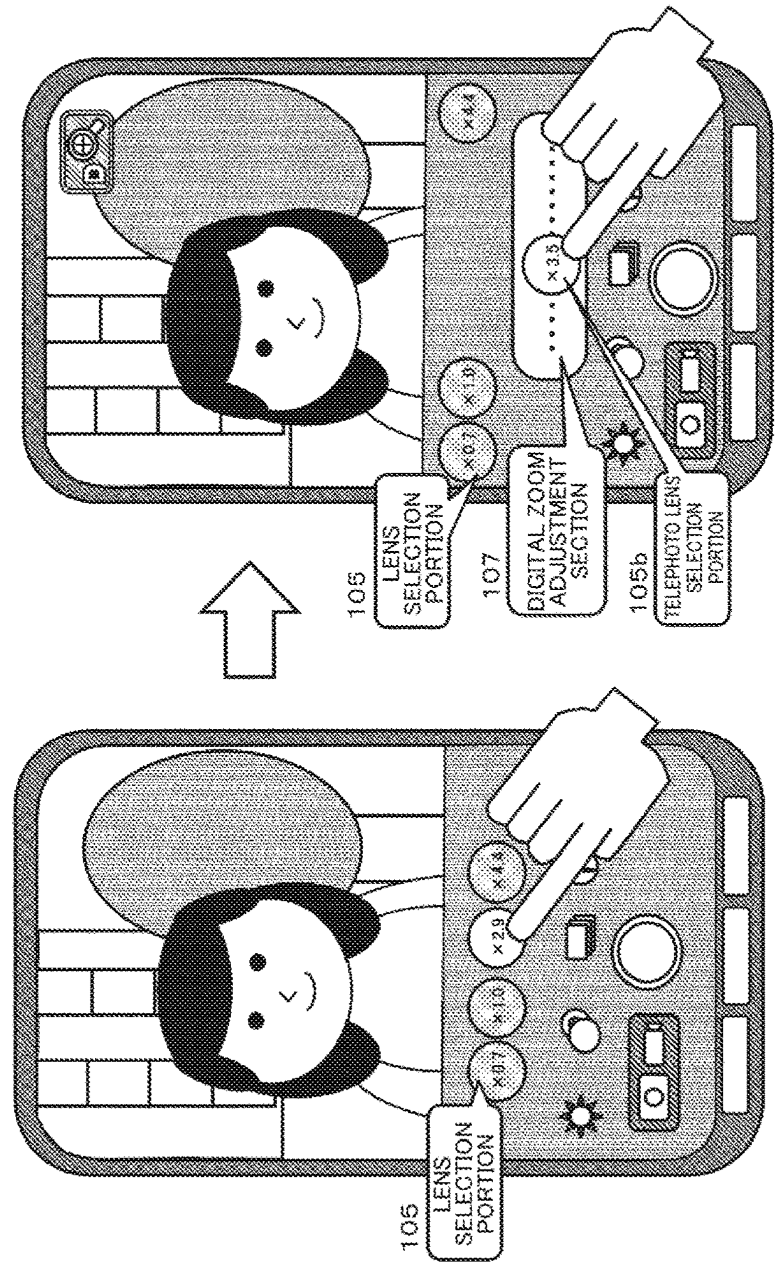

105
LENS SELECTION PORTION
107
DIGITAL ZOOM ADJUSTMENT SECTION
105b
TELEPHOTO LENS SELECTION PORTION
×3.5
105
LENS SELECTION PORTION
×2.9

Fig. 27

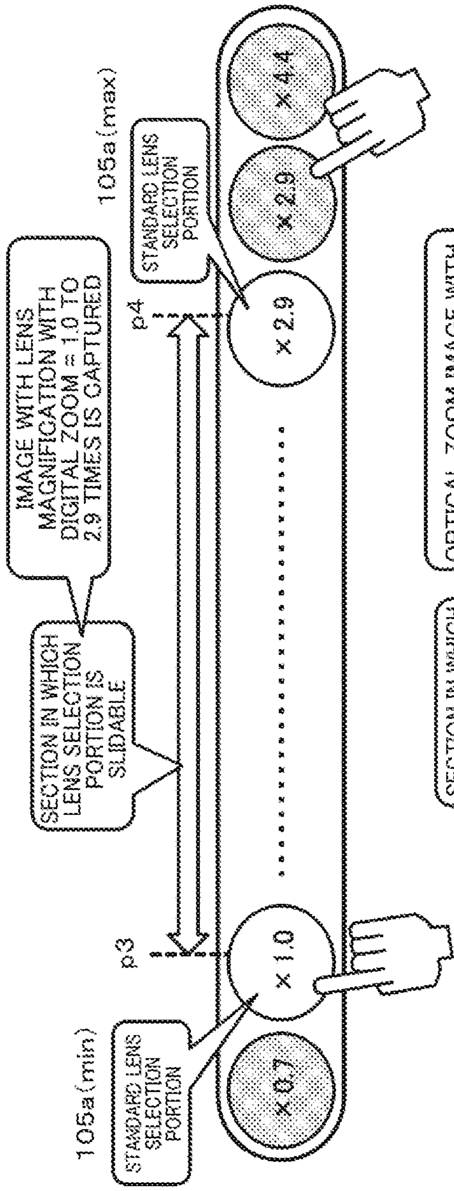



Fig. 28A

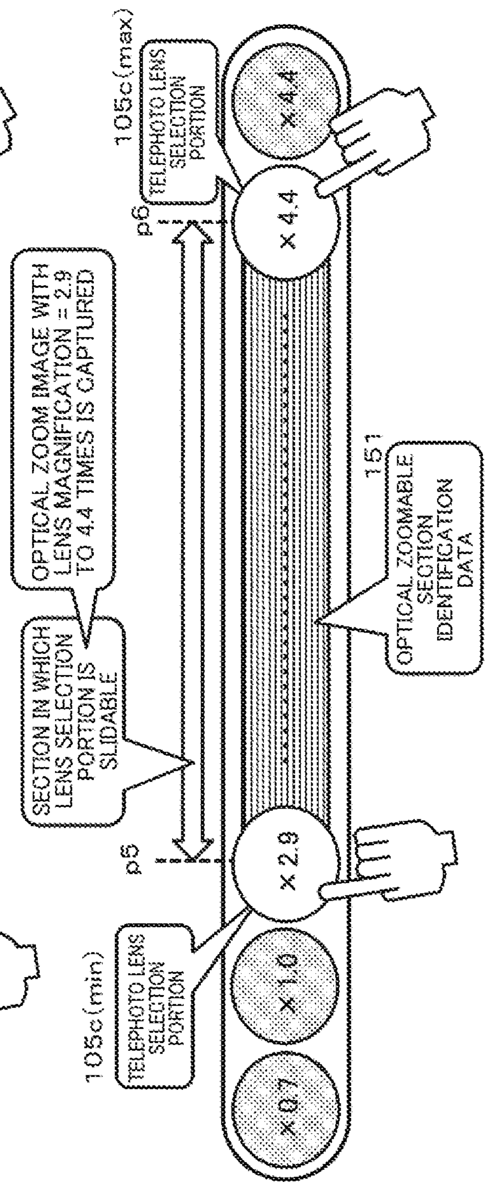



Fig. 28B

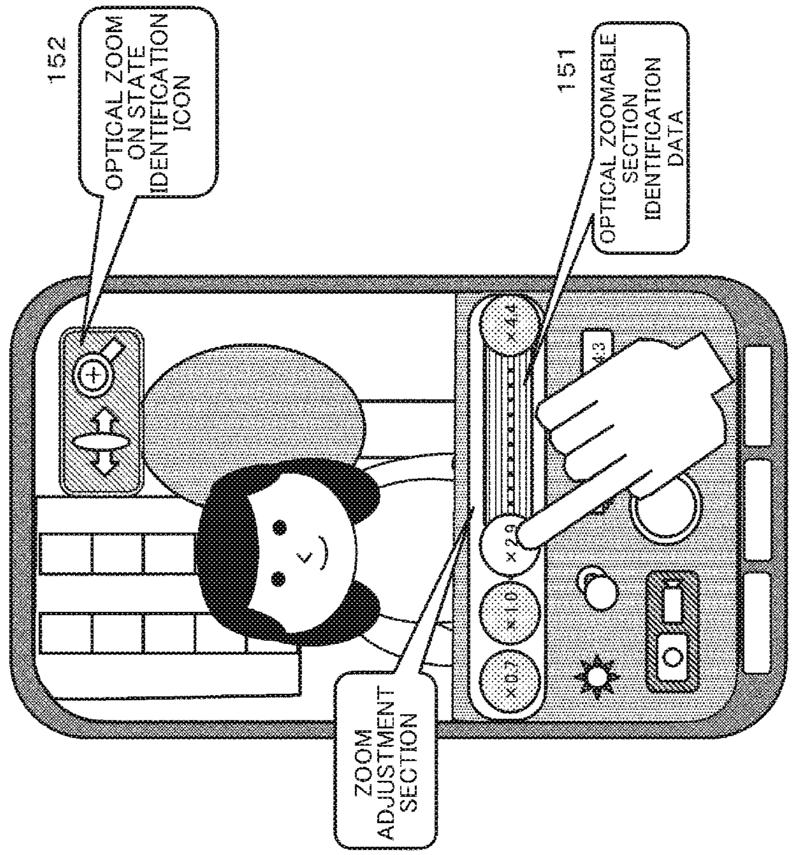

152
OPTICAL ZOOM ON STATE IDENTIFICATION ICON
151
OPTICAL ZOOMABLE SECTION IDENTIFICATION DATA
ZOOM ADJUSTMENT SECTION
×0.7
×1.0
×2.9
×4.4

Fig. 29

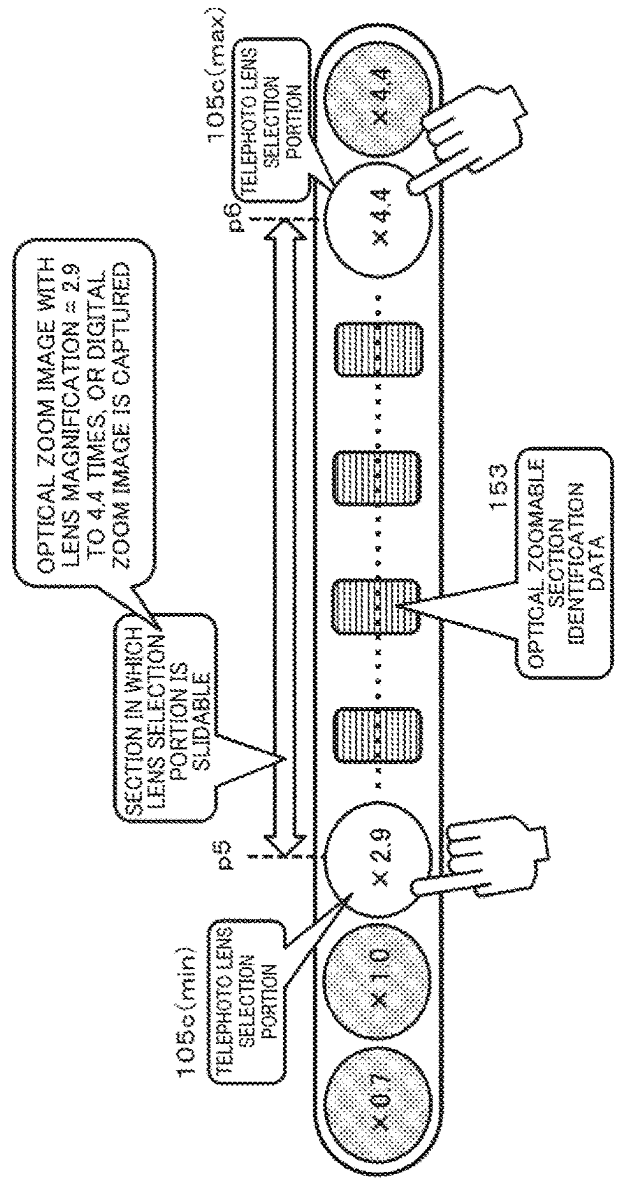
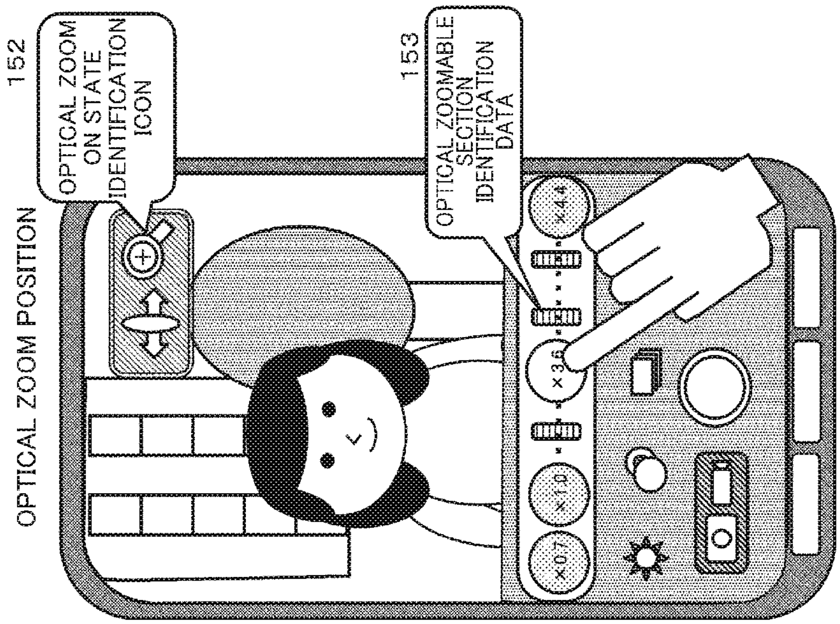

105c(max)
TELEPHOTO LENS SELECTION PORTION
×4.4
×4.4
p6
OPTICAL ZOOM IMAGE WITH LENS MAGNIFICATION = 2.9 TO 4.4 TIMES, OR DIGITAL ZOOM IMAGE IS CAPTURED
SECTION IN WHICH LENS SELECTION PORTION IS SLIDABLE
153
OPTICAL ZOOMABLE SECTION IDENTIFICATION DATA
p5
×2.9
105c(min)
TELEPHOTO LENS SELECTION PORTION
×1.0
×0.7

Fig. 30

Fig. 31B



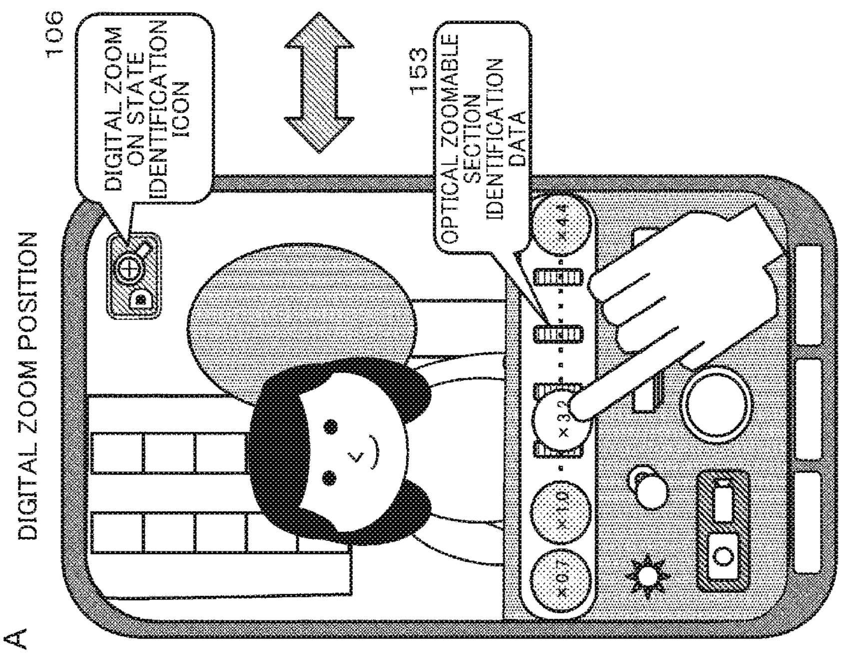




Fig. 31A

Fig. 32A

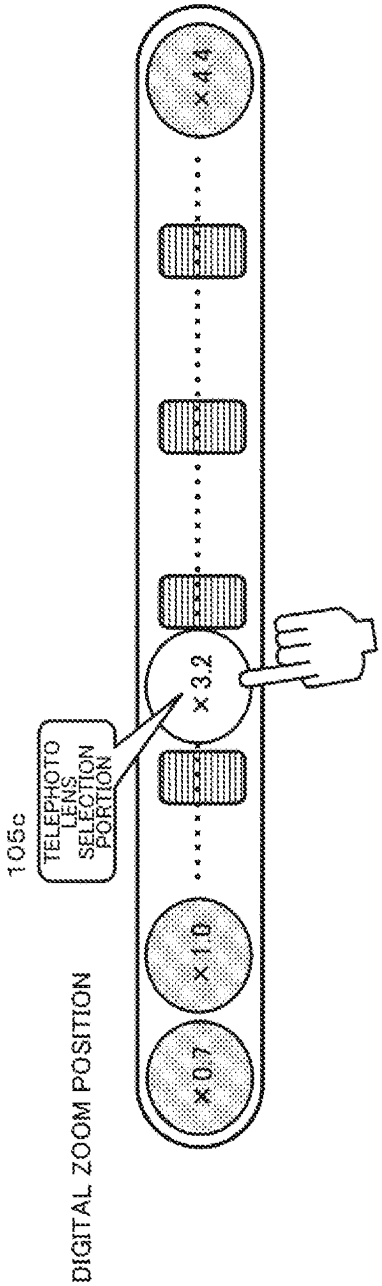



Fig. 32B

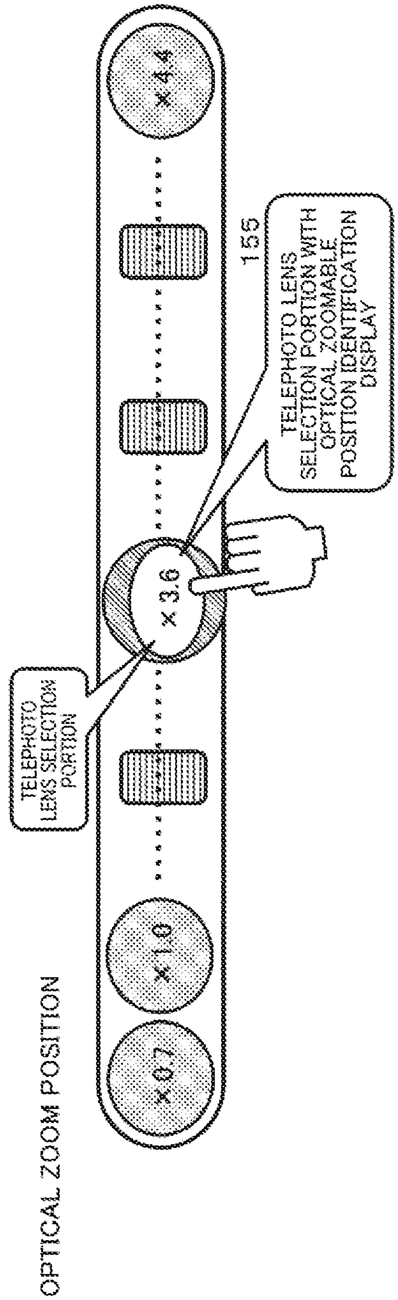
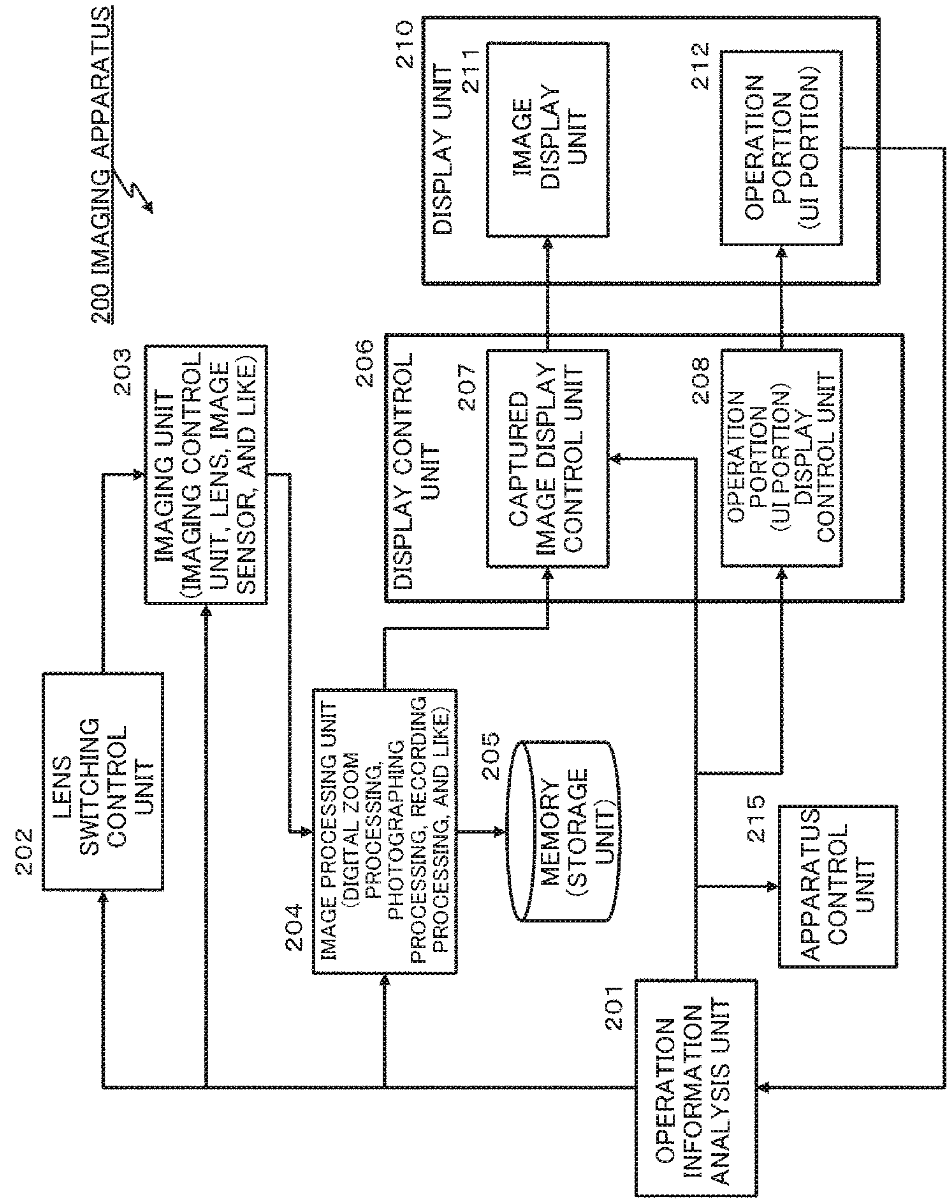
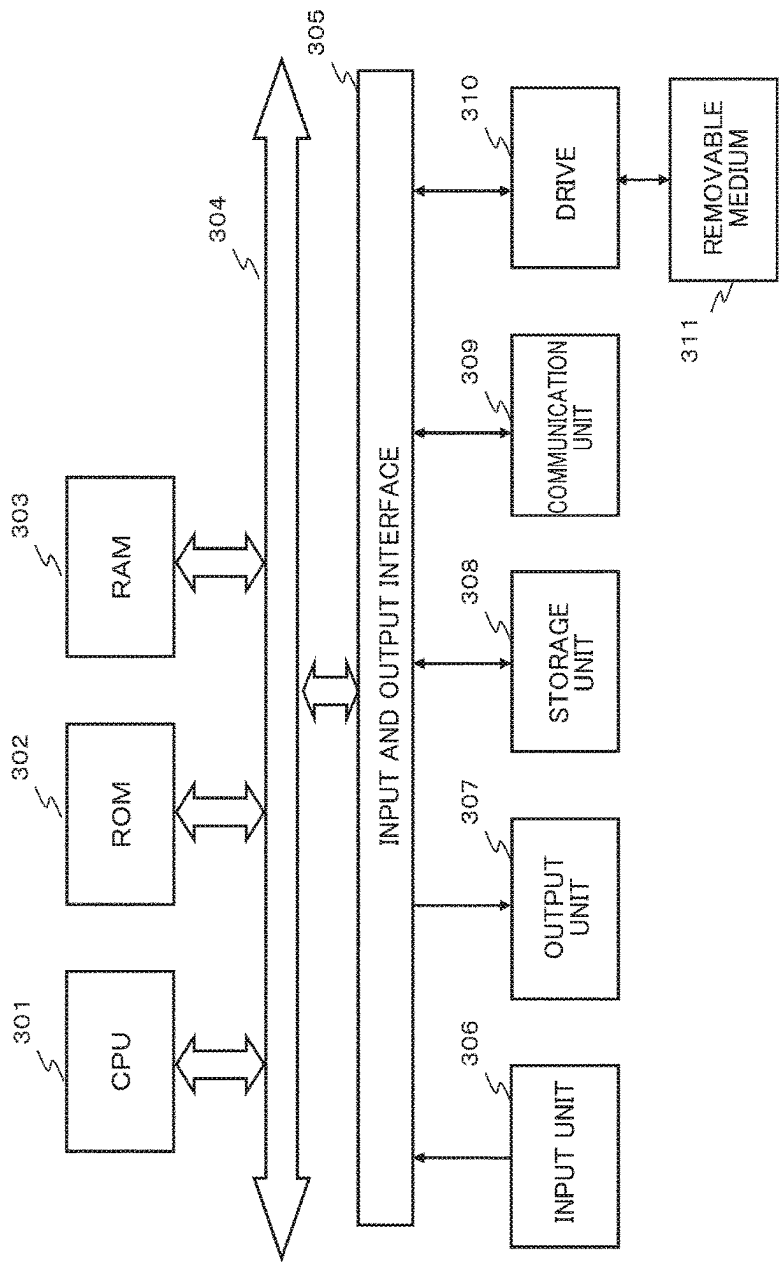


Fig. 33


200 IMAGING APPARATUS
201
OPERATION INFORMATION ANALYSIS UNIT
202
LENS SWITCHING CONTROL UNIT
203
IMAGING UNIT (IMAGING CONTROL UNIT, LENS, IMAGE SENSOR, AND LIKE)
204
IMAGE PROCESSING UNIT (DIGITAL ZOOM PROCESSING, PHOTOGRAPHING PROCESSING, RECORDING PROCESSING, AND LIKE)
205
MEMORY (STORAGE UNIT)
206
DISPLAY CONTROL UNIT
207
CAPTURED IMAGE DISPLAY CONTROL UNIT
208
OPERATION PORTION (UI PORTION) DISPLAY CONTROL UNIT
210
DISPLAY UNIT
211
IMAGE DISPLAY UNIT
212
OPERATION PORTION (UI PORTION)
215
APPARATUS CONTROL UNIT



301
CPU
302
ROM
303
RAM
304
305
INPUT AND OUTPUT INTERFACE
306
INPUT UNIT
307
OUTPUT UNIT
308
STORAGE UNIT
309
COMMUNICATION UNIT
310
DRIVE
311
REMOVABLE MEDIUM


Fig. 34

# IMAGING APPARATUS, IMAGING APPARATUS CONTROL METHOD, AND PROGRAM

## CROSS REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase of International Patent Application No. PCT/JP2022/002357 filed on Jan. 24, 2022, which claims priority benefit of Japanese Patent Application No. JP 2021-054027 filed in the Japan Patent Office on Mar. 26, 2021. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

## TECHNICAL FIELD

The present disclosure relates to an imaging apparatus, an imaging apparatus control method, and a program. More specifically, the present invention relates to an imaging apparatus that improves operability of switching between lenses used in the imaging apparatus and digital zoom processing, an imaging apparatus control method, and a program.

## BACKGROUND ART

For example, many portable electronic devices such as smartphones have a camera function.

Especially in recent years, there has been an increase in the number of smartphones with a multi-lens camera including a plurality of different lenses, such as a wide-angle lens and a telephoto lens, in addition to a standard lens.

In the multi-lens camera, the lenses are switched therebetween depending on, for example, a distance to a subject and are used so that clearer images of both a distant subject and a nearby subject can be captured.

However, each of a standard lens, a wide-angle lens, and a telephoto lens mounted on a smartphone is a single focal length lens with a fixed focal length. Further, many thin apparatuses such as smartphones do not have a zoom lens. Therefore, for example, digital zoom processing is performed when an intermediate image between an image captured using a standard lens and an image captured using a telephoto lens is generated.

Digital zoom processing is performed as image enlargement processing for a captured image. For example, a portion of the image captured using the standard lens is cropped and enlarged so that it is possible to generate an intermediate image between an image captured using the standard lens and an image captured using the telephoto lens.

Digital zoom processing is described in, for example, PTL 1 (JP 2013-175855A).

When an image is captured using the smartphone including the multi-lens camera, a user first selects one of lenses to be used for capturing.

Through this lens selection processing, an image captured by the selected lens, that is, a so-called through image is displayed on a display unit of the smartphone. Further, when an angle of view or size of the image displayed on the display unit is adjusted, the user performs zoom processing using a digital zoom function, displays the image corrected by the digital zooming on the display unit, and sets the angle of view or size of the image to a desired value.

Thereafter, the user operates the shutter to perform photographing.



Thus, when photographing using the multi-lens camera is performed, the user is required to sequentially execute two processing including the lens selection processing and digital zoom processing.

However, in the case of most user interfaces (UIs) for a display of an operation portion of a smartphone, a lens selection UI and a digital zoom processing UI are configured as separate UIs. Therefore, when the user tries to perform digital zooming after selecting a lens, it is necessary to switch between the UIs, and there has often been a problem in that the user misses a photographing timing during this time.

## CITATION LIST

### Patent Literature

[PTL 1]
JP 2013-175855A

## SUMMARY

### Technical Problem

The present disclosure has been made, for example, in view of the above problems, and an object of the present disclosure is to provide an imaging apparatus in which the operability of switching of lenses used in the imaging apparatus, such as a smartphone, and digital zoom processing is improved, an imaging apparatus control method, and a program.

### Solution to Problem

A first aspect of the present disclosure is an imaging apparatus including:

a plurality of lenses available for image capture; and

a display control unit configured to execute control of display data to be output to a display unit,

wherein the display control unit

displays a plurality of lens selection portions corresponding to the plurality of respective lenses on a display unit, and

displays a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

Further, a second aspect of the present disclosure is

an imaging apparatus control method executed in an imaging apparatus,

wherein the imaging apparatus includes

a plurality of lenses available for image capture; and

a display control unit configured to execute control of display data to be output to a display unit,

wherein the display control unit

displays a plurality of lens selection portions corresponding to the plurality of respective lenses on a display unit, and

displays a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

Further, a third aspect of the present disclosure is

a program for causing imaging apparatus control processing to be executed in an imaging apparatus,

a plurality of lenses available for image capture; and

a display control unit configured to execute control of display data to be output to a display unit, and

the program causes the display control unit to execute

processing for displaying a plurality of lens selection portions corresponding to the plurality of respective lenses on a display unit, and

processing for displaying a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

The program of the present disclosure is, for example, a program that can be provided to an information processing apparatus or a computer system capable of executing various program codes, in a computer-readable format by a storage medium or communication medium. When such a program is provided in a computer-readable format, processing according to the program is realized on the information processing apparatus or computer system.

Still other objects, characteristics, and advantages of the present disclosure will become apparent from more detailed description based on embodiments of the present disclosure or the accompanying drawings, which will be described below. In the present specification, the system is a logical collective configuration of a plurality of devices, and the devices of the respective configuration are not limited to being in the same housing.

According to the configuration of the embodiment of the present disclosure, the operation portion (UI) that enables efficient execution of the photographing lens selection processing by the user and the digital zoom processing for the captured image using the selected lens is realized.

Specifically, for example, a plurality of lenses with different lens magnifications available for image capturing, and a display control unit configured to execute control of display data to be output to a display unit are included, and the display control unit displays a plurality of lens selection portions corresponding to the plurality of respective lenses on the display unit, displays a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit, and displays a digital zoom image executed depending on the slide amount on the display unit.

With this configuration, the operation portion (UI) that enables efficient execution of the photographing lens selection processing by the user and the digital zoom processing for the captured image using the selected lens is realized.

The effects described in the present specification are merely examples and are not limited, and there may be additional effects.

## BRIEF DESCRIPTION OF DRAWINGS

FIGS. **1**A and **1**B are diagrams illustrating an example of an external appearance configuration of an imaging apparatus.

FIGS. **2**A, **2**B, **2**C, and **2**D are diagrams illustrating an example of an image captured by lens switching.

FIG. **3** is a diagram illustrating an example of display data on a display unit of the imaging apparatus.

FIGS. **4**A, **4**B, **4**C, and **4**D are diagrams illustrating an example of a captured image and display data according to a user operation with respect to a lens selection portion (lens selection icon).

FIG. **5** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **6** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **7** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **8** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **9** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **10** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **11** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **12** is a diagram illustrating an example of a series of operations by a user who operates the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **13** is a diagram illustrating an example of a series of operations by the user who operates the lens selection portion (lens selection icon) of the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **14** is a diagram illustrating an example of a series of operations by the user who operates the lens selection portion (lens selection icon) of the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **15** is a diagram illustrating an example of a series of operations by the user who operates the lens selection portion (lens selection icon) of the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **16** is a diagram illustrating an example of a series of operations by the user who operates the lens selection portion (lens selection icon) of the imaging apparatus and an example of transition of display data according to the user operation.

FIG. **17** is a diagram illustrating a flowchart illustrating a processing sequence that is executed by the imaging apparatus of the present disclosure.

FIG. **18** is a diagram illustrating a specific example of image enlargement processing in digital zoom processing.

FIG. **19** is a diagram illustrating a specific example of image quality degradation in the digital zoom processing.

FIGS. **20**A, **20**B, and **20**C are diagrams illustrating a specific example of display data for notifying the user of deterioration of image quality due to the digital zoom processing.

FIGS. **21**A, **21**B, and **12**C are diagrams illustrating a specific example of display data for notifying the user of deterioration of image quality due to the digital zoom processing.

FIGS. **22**A, **22**B, and **22**C are diagrams illustrating a specific example of display data for notifying the user of deterioration of image quality due to the digital zoom processing.

FIGS. **23**A, **23**B, and **23**C are diagrams illustrating a specific example of notifying the user of deterioration of image quality due to digital zoom processing.

FIGS. **24**A, **24**B, and **24**C are diagrams illustrating a specific example of notifying the user of deterioration of image quality due to digital zoom processing.

FIG. **25** is a diagram illustrating an example of the display data of an operation portion (UI portion).

FIG. **26** is a diagram illustrating an example of the display data of the operation portion (UI portion).

FIG. **27** is a diagram illustrating an example of the display data of the operation portion (UI portion).

FIGS. **28**A and **28**B are diagrams illustrating an example of display data of an example unit (UI portion) in an apparatus capable of optical zooming.

FIG. **29** is a diagram illustrating an example of display data of an example unit (UI portion) in an apparatus capable of optical zooming.

FIG. **30** is a diagram illustrating an example of display data of an example unit (UI portion) in an apparatus capable of optical zooming.

FIGS. **31**A and **31**B are diagrams illustrating an example of display data of an example unit (UI portion) in an apparatus capable of optical zooming.

FIGS. **32**A and **32**B are diagrams illustrating an example of display data of an example unit (UI portion) in an apparatus capable of optical zooming.

FIG. **33** is a diagram illustrating a configuration example of the imaging apparatus of the present disclosure.

FIG. **34** is a diagram illustrating a hardware configuration example of the imaging apparatus of the present disclosure.

## DESCRIPTION OF EMBODIMENTS

Hereinafter, details of the imaging apparatus, the imaging apparatus control method, and the program according to the present disclosure will be described with reference to the drawings. Description will be made according to the following items.

1. Overview of Configuration of Imaging Apparatus of Present Disclosure and Lens Switching Processing
2. Operation Portion Display Control Processing Executed in Imaging Apparatus of Present Disclosure
3. Operation Portion UI Display Control Processing Sequence Executed by Imaging Apparatus of Present Disclosure
4. Display Control Processing for Notifying User of Image Quality Degradation Due to Digital Zoom
5. Other Examples of Display Control Processing
6. Configuration Example of Imaging Apparatus of Present Disclosure
7. Hardware Configuration Example of Imaging Apparatus
8. Conclusion of Configuration of Present Disclosure

## 1. OVERVIEW OF CONFIGURATION OF IMAGING APPARATUS OF PRESENT DISCLOSURE AND LENS SWITCHING PROCESSING

First, an overview of a configuration of the imaging apparatus of the present disclosure and lens switching processing will be described.

FIGS. **1**A and **1**B are diagrams illustrating a configuration example of an imaging apparatus of the present disclosure. In FIGS. **1**A and **1**B, a smartphone having a camera function is illustrated as an example of an imaging apparatus **10** of the present disclosure.

The imaging apparatus of the present disclosure is not limited to such a smartphone, and includes, for example, a tablet terminal having a camera function, a PC, and an imaging apparatus such as a camera.

FIG. **1**A illustrates a front side of the imaging apparatus (smartphone) **10**, in which a display unit **11** is provided. FIG. **1**B illustrates a back side of the imaging apparatus (smartphone) **10**, in which a camera lens **12** is provided.

The camera lens **12** is configured of a plurality of lenses with different focal lengths. That is, the imaging apparatus **10** illustrated in FIGS. **1**A and **1**B are smartphones including the multi-lens camera.

The user can select a lens to be used from among the plurality of lenses at the time of capturing an image, and appropriately perform switching processing to clearly photograph subjects at various distances.

The imaging apparatus (smartphone) **10** illustrated in FIGS. **1**A and **1**B include the following four types of lenses.

(a) Standard lens **12***a*

(b) Wide-angle lens **12***b*

(c) Telephoto lens **12***c*

(d) Super-telephoto lens **12***d*

In the figure, the telephoto lens **12***c* and the super-telephoto lens **12***d* are shown in one lens portion at a bottom of the camera lens **12**, but two lenses including the telephoto lens **12***c* and the super-telephoto lens **12***d* are incorporated in this lens portion at the bottom.

(a) The standard lens **12***a* is a lens with a focal length of 24 mm and a lens magnification of 1.0 times.

(b) The wide-angle lens **12***b* is a lens with a focal length of 16 mm and a lens magnification of 0.7 times.

(c) The telephoto lens **12***c* is a lens with a focal length of 70 mm and a lens magnification of 2.9 times.

(d) The super-telephoto lens **12***d* is a lens with a focal length of 105 mm and a lens magnification of 4.4 times.

A user of the imaging apparatus **10** can switch between these lenses **12***a* to **12***d* to perform photographing at the time of image capturing.

The focal length and the lens magnification of each lens are examples, and the processing of the present disclosure can also be applied to an apparatus configuration including various lenses having other focal lengths and lens magnifications.

Further, the lens type is also an example, and the lens type is not limited to four types, and the configuration of the present disclosure is also applicable to configurations including two, three, four, five, six, and more types of lenses.

FIGS. **2**A, **2**B, **2**C, and **2**D illustrate an example of images captured by switching between the lenses FIGS. **2**A, **2**B, **2**C, and **2**D.

In FIGS. **2**A, **2**B, **2**C, and **2**D, examples of the following four types of captured images are illustrated.

FIG. **2**A Image captured using standard lens

FIG. **2**B Image captured using wide-angle lens

FIG. **2**C Image captured using telephoto lens

FIG. **2**D Image captured using super-telephoto lens

FIG. **2**A The image captured using the standard lens is an image captured using the standard lens **12***a* with a focal length of 24 mm and a lens magnification of 1.0 times.

FIG. **2**B The image captured using the wide-angle lens is an image captured using the wide-angle lens **12***b* with a focal length of 16 mm and a lens magnification of 0.7 times.

FIG. **2**C The image captured using the telephoto lens is an image captured using a telephoto lens **12***c* with a focal length of 70 mm and a lens magnification of 2.9 times.

FIG. **2**D The image captured using super-telephoto lens is an image captured using the super-telephoto lens **12***d* with a focal length of 105 mm and a lens magnification of 4.4 times.

As can be understood from FIGS. **2**A, **2**B, **2**C, and **2**D, FIG. **2**B the image captured using the wide-angle lens is an image captured using the widest field of view, and an order of the width of fields of view is an order of FIG. **2**B the image captured using the wide-angle lens, FIG. **2**A the image captured using the standard lens, FIG. **2**C the image captured using the telephoto lens, and FIG. **2**D the image captured using the super-telephoto lens.

Meanwhile, when sizes of subjects at the same distance at a distance are compared with each other, FIG. **2**D the image captured using the super-telephoto lens is the largest. The order of sizes of the subject at the same distance is reverse to order of fields of view, as in FIG. **2**D the image captured using the super-telephoto lens, FIG. **2**C the image captured using the telephoto lens, FIG. **2**A the image captured using the standard lens, and FIG. **2**B the image captured using the wide-angle lens.

Thus, with the multi-lens camera, it is possible to capture various images by switching between lenses to be used.

## 2. OPERATION PORTION DISPLAY CONTROL PROCESSING EXECUTED IN IMAGING APPARATUS OF PRESENT DISCLOSURE

Next, operation portion display control processing executed in the imaging apparatus of the present disclosure will be described.

FIG. **3** is a diagram illustrating an example of an operation portion serving as a user interface (UI) displayed on the display unit **11** of the imaging apparatus (smartphone) **10**.

An upper part of the display unit **11** is set as a captured image display area **101**, and a lower part thereof is set as an operation portion display area **102**.

The image displayed in the captured image display area **101** is not limited to an image subjected to capturing processing and also includes an image before capturing. A so-called through image, that is, an image that the user is about to capture is also displayed in the captured image display area **101**. The user operates operation portions such as various icons displayed in the operation portion display area **102** to capture an image while viewing the image displayed in the captured image display area **101**.

In the operation portion display area **102** illustrated in FIG. **3**, icons as various operation portions such as a shutter **103**, a still image/moving image switching portion **104**, and a lens selection portion **105** are displayed. Information on the operation portions (icons) displayed in the operation portion display area **102** can be switched to various settings.

When an image displayed in the captured image display area **101** is captured, the user touches the shutter **103**. Through this processing, the image displayed in the captured image display area **101** is captured and recorded in a memory.

The still image/moving image switching portion **104** is an operation portion for switching between a still image and a moving image as a captured image.

The processing of the present disclosure can be applied to both still image capturing processing and moving image capturing processing. However, in the following embodiments, a processing example in the case of capturing a still image will be described as a representative example.

The lens selection portion **105** is an operation portion for switching between the lenses that are used for image capturing, and four lens selection icons for individually selecting the four types of lenses described above with reference to FIGS. **1**A and **1**B are displayed.

FIG. **3** illustrates the following four lens selection portions (lens selection icons).

Wide-angle lens selection portion (×0.7)

Standard lens selection portion (×1.0)

Telephoto lens selection portion (×2.9)

Super-telephoto lens selection portion (×4.4)

Numerical values (0.7 to 4.4) within the respective icons indicate the magnifications of the respective lenses.

The user can perform switching to a lens to be used, by operating (touching) one of the four lens selection portions (lens selection icons).

FIGS. **4**A, **4**B, **4**C, and **4**D are diagrams illustrating an example of display images switched depending on a user operation (touch) with respect to the four lens selection portions (lens selection icons), that is, an example of switching between the display images in the captured image display area **101**.

FIGS. **4**A, **4**B, **4**C, and **4**D illustrate examples of the following four types of user operation states.

FIG. **4**A A state in which the user operates (touches) the standard lens selection portion (standard lens selection icon)

FIG. **4**B A state in which the user operates (touches) the wide-angle lens selection portion (wide-angle lens selection icon)

FIG. **4**C A state in which the user operates (touches) the telephoto lens selection portion (telephoto lens selection icon)

FIG. **4**D A state in which the user operates (touches) the super-telephoto lens selection portion (super-telephoto lens selection icon)

As illustrated in FIG. **4**A, when the user operates (touches) the standard lens selection portion (standard lens selection icon), processing of switching a photographing lens to the standard lens **12***a* is executed inside the imaging apparatus **10**, and the image (through image) captured using the standard lens **12***a* is displayed on the display unit **11**.

Further, as illustrated in FIG. **4**B, when the user operates (touches) the wide-angle lens selection portion (wide-angle lens selection icon), processing of switching the photographing lens to the wide-angle lens **12***b* is executed inside the imaging apparatus **10**, and the image (through image) captured using the wide-angle lens **12***b* is displayed on the display unit **11**.

Further, as illustrated in FIG. **4**C, when the user operates (touches) the telephoto lens selection portion (telephoto lens selection icon), processing of switching the photographing lens to the telephoto lens **12***c* is executed inside the imaging apparatus **10**, and the image (through image) captured using the telephoto lens **12***c* is displayed on the display unit **11**.

Further, as illustrated in FIG. **4**D, when the user operates (touches) the super-telephoto lens selection portion (super-telephoto lens selection icon), processing of switching the photographing lens to the super-telephoto lens **12***d* is executed inside the imaging apparatus **10**, and the image (through image) captured using the super-telephoto lens **12***d* is displayed on the display unit **11**.

Thus, the user performs an operation (touch) for selecting one lens selection icon from among the lens selection portions (lens selection icons) **105** in the operation portion (UI), so that it becomes possible to display the image captured using the selected lens on the display unit **11**.

Then, the shutter **103** is touched so that it is possible to capture the image displayed on the display unit **11** and record the image in the memory.

However, images captured by selecting the respective lens selection portions (lens selection icons) **105** in the operation portion (UI) are only four types of images according to four types of magnifications such as magnifications (0.7 times, 1.0 times, 2.9 times, and 4.4 times) of the respective lenses.

For example, the image captured using the standard lens illustrated in FIG. **4**A is an image captured according to the lens magnification (1.0 times) unique to the standard lens.

Further, the image captured using the telephoto lens illustrated in FIG. **4**C is an image captured according to a lens magnification (2.9 times) unique to the telephoto lens.

When it is desired to capture an intermediate image between the image captured using the standard lens illustrated in FIG. **4**A and the image captured using the telephoto lens illustrated in FIG. **4**C, it is necessary to perform digital zoom processing.

Digital zoom processing is performed as image enlargement processing for a captured image. For example, a portion of the image captured using the standard lens is cropped and enlarged so that it is possible to generate an intermediate image between the image captured using the standard lens and the image captured using the telephoto lens.

A specific example of a user operation when digital zoom processing is performed in the imaging apparatus **10** of the present disclosure will be described below.

FIGS. **5** to **12** illustrate a series of operation examples of the user who operates the imaging apparatus **10**.

Depending on the user operation, display states of the display unit **11** of the imaging apparatus **10** sequentially transition from (step S**1**) in FIG. **5** to (step S**10**) in FIG. **12**.

Hereinafter, display data on the display unit **11** and a user operation in each step will be described.

(Step S**1**)

The display data shown in step S**1** of FIG. **5** is an initial screen when the camera is started.

The standard lens **12***a* is selected by default when the camera is started.

An image captured using the standard lens **12** is displayed in the captured image display area **101**.

In the lens selection portion (lens selection icon) **105** of the operation portion display area **102**, only a standard lens selection icon (1.0) is displayed with higher brightness or different color than or from other lens selection icons (0.7, 2.9, and 4.4) to show that the standard lens **12***a* is selected.

When the user confirms the image displayed in the captured image display area **101**, that is, the image captured using the standard lens **12** with a lens magnification of 1.0 times, and determines to capture and record this image, the user touches the shutter **103**.

However, for example, when a person is desired to be photographed a little larger, digital zoom is performed. Processing in this case is (step S**2**) and subsequent processing.

(Step S**2**)

Step S**2** in FIG. **5** is processing when the user wants to photograph a person a little larger and starts digital zooming.

In this case, the user touches the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times, which is a lens selection portion (lens selection icon) of a currently selected lens, and performs processing for sliding this icon.

A specific sequence of lens selection icon slide processing will be described with reference to FIG. **6** and subsequent figures.

(Step S**3**)

When the user operates (touches) the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times, the data processing unit inside the imaging apparatus **10** displays a digital zoom ON state identification icon **106** and a digital zoom adjustment section **107**, as illustrated in FIG. **6**, according to icon touch processing.

The digital zoom ON state identification icon **106** is an icon for notifying the user of a setting indicating that the digital zoom processing is possible.

The digital zoom adjustment section **107** is an area in which the lens selection icon is slidable, and is a section in which a digital zoom amount can be set.

The user can set the digital zoom amount by sliding the touched lens selection icon within the digital zoom adjustment section **107**.

The user touches the lens selection icon for the standard lens **12***a* with lens magnification of 1.0 times and slides the lens selection icon for the standard lens **12***a* within the digital zoom adjustment section. With this slide processing, it is possible to adjust the digital zoom amount for the image captured using the standard lens **12***a* with a lens magnification of 1.0 times, which is the selected lens. Processing for adjusting this digital zoom amount will be described with reference to FIG. **7** and subsequent drawings.

(Step S**4**)

FIG. **7** illustrates display data of (step S**3**) described with reference to FIG. **6**, and display data of subsequent (step S**4**).

In the display data of (step S**3**), the digital zoom ON state identification icon **106** and the digital zoom adjustment section **107** are displayed in response to the user touching the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times.

Thereafter, the user slides the touched lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times to the right within the digital zoom adjustment section **107**.

That is, the lens selection icon for the standard lens **12***a* with a lens magnification of 1.0 times is slid (moved) toward a lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times.

According to this slide processing, the numerical value displayed on the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times is changed.

In (step S**4**) of FIG. **7**, the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times touched by the user is slid to the right, and (×1.4) is displayed on the icon at this position.

This indicates that digital zoom processing for enlarging the image captured using the standard lens **12***a* with a lens magnification of 1.0 times to an image corresponding to an image captured using a lens with a lens magnification of 1.4 times is performed.

A digital zoom image of the image captured using the standard lens **12***a* having a lens magnification of 1.0 times touched by the user is displayed in the captured image display area **101**. That is, an image corresponding to the image captured using the lens with a lens magnification of 1.4 times is displayed.

The data processing unit of the imaging apparatus executes image correction processing for the image captured by the standard lens **12***a* with the lens magnification of 1.0 times depending on a slide amount of the lens selection icon slid by the user. The data processing unit of the imaging apparatus executes processing for correcting the captured image, that is, processing for extracting a partial area from the captured image and processing for enlarging an extracted image as digital zoom processing, and displays an image after the enlarging processing in the captured image display area **101**.

The image displayed in the captured image display area of (step S**4**) in FIG. **7** is an image subjected to this digital zoom processing.

That is, the image is an image generated by executing the digital zoom processing for extracting a partial area from the image captured by the standard lens **12***a* with a lens magnification of 1.0 times, which is the image displayed in the captured image display area (step S**3**) in FIG. **7** and enlarging the extracted image. This image corresponds to the image captured using the lens with the lens magnification of 1.4 times.

When the user confirms the image displayed in the captured image display area **101**, that is, the same digital zoom image as the image captured using the lens with the lens magnification of 1.4 times and determines that the user desires to record this image, the user touches the shutter **103**.

However, for example, when a person is desired to be photographed a little larger, digital zoom is also performed. Processing in this case is (step S**5**) and subsequent processing.

(Step S**5**)

The display data illustrated in FIG. **8** shows a state in which the user has slid the lens selection icon for the standard lens **12***a* with a lens magnification of 1.0 times to the right within the digital zoom adjustment section **107**.

That is, the lens selection icon for the standard lens **12***a* with a lens magnification of 1.0 times is slid (moved) to be closer to the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times.

According to this slide processing, the numerical value displayed on the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times is further changed.

In (step S**5**) of FIG. **8**, the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times touched by the user is further slid to the right, and (×2.9) is displayed on the icon at this position.

This indicates that digital zoom processing for enlarging the image captured using the standard lens **12***a* with a lens magnification of 1.0 times to an image corresponding to an image captured using a lens with a lens magnification of 2.9 times is performed.

A digital zoom image of the image captured using the standard lens **12***a* having a lens magnification of 1.0 times touched by the user is displayed in the captured image display area **101**. That is, an image corresponding to an image captured using the lens with the lens magnification of 2.9 times is displayed.

As described above, the data processing unit of the imaging apparatus executes image correction processing for the image captured by the standard lens **12***a* with the lens magnification of 1.0 times depending on a slide amount of the lens selection icon slid by the user.

In (step S**5**) of FIG. **8**, a slide amount of the lens selection icon is larger than that of (step S**4**) in FIG. **7**, and the data processing unit of the imaging apparatus performs digital zoom processing according to the slide amount. Here, an enlarged image corresponding to the image captured using the lens with the lens magnification of 2.9 times is generated.

The image displayed in the captured image display area of (step S**5**) in FIG. **8** is the image subjected to this digital zoom processing, and corresponds to the image captured using the lens with the lens magnification of 2.9 times.

(Step S**6**)

(Step S**9**) illustrated in FIG. **9** is a diagram illustrating a user operation when a determination is made that the user desires to record the image displayed in the captured image display area **101**.

The image displayed in the captured image display area **101** of the display unit **11** of the imaging apparatus in (step S**9**) illustrated in FIG. **9** is the image displayed in (step S**5**) described with reference to FIG. **8**.

That is, this is the digital zoom processing image for the image captured using the standard lens with a lens magnification of 1.0 times. This image is the same as the image captured using the lens with the lens magnification of 2.9 times.

The user confirms this displayed image and touches the shutter **103** when the user determines that the user desires to record this image.

Through this touch processing, processing for capturing the image displayed in the captured image display area **101** is executed, and the image is recorded in the memory.

The image recorded in the memory is the digital zoom processing image of the image captured using the standard lens with a lens magnification of 1.0 times.

Further, when the user wants to capture an image in which a face of the person is larger, the user performs lens switching.

As illustrated in FIG. **9**, the lens selection icon selected and slid by the user is in contact with an adjacent lens selection icon (×2.9) with a lens magnification of 2.9 times within the digital zoom adjustment section **107**, and cannot further slide to the right.

Therefore, when the user is desired to capture the image in which a face of the person is larger, the user performs lens switching.

This processing will be described with reference to FIG. **10**.

(Step S**7**)

When the user captures the digital zoom processing image (an image corresponding to the lens magnification of 2.9 times) for the image captured using the standard lens with a lens magnification of 1.0 times in (step S**6**) illustrated in FIG. **9**, and then, desires to capture the image in which a face of the person is larger, the user performs lens switching.

As illustrated in FIG. **10** (step S**7**), the user takes away the finger from the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times that has been subjected to the slide processing, and touches the adjacent lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times.

When the user takes away the finger from the lens selection icon for the standard lens **12***a* with the lens magnification of 1.0 times that has been subjected to the slide processing, the digital zoom ON adjustment icon **106** and the digital zoom adjustment section **107** are not displayed.

Thereafter, the user touches the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times.

(Step S**8**)

The display data of (step S**8**) illustrated in FIG. **10** is display data after the user touches the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times.

When the user touches the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times, the data processing unit inside the imaging apparatus **10** displays the digital zoom ON state identification icon **106** and the digital zoom adjustment section **107**, as illustrated in FIG. **10** (step S**8**), according to detection of the icon touch processing.

The digital zoom ON state identification icon **106** is an icon for notifying the user of a setting indicating that the digital zoom processing is possible.

The digital zoom adjustment section **107** is an area in which the lens selection icon is slidable, and is a section in which the digital zoom amount can be set.

The user can set the digital zoom amount by sliding the touched lens selection icon within the digital zoom adjustment section **107**.

The user touches the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times and slides the lens selection icon within the digital zoom adjustment section. With this slide processing, it is possible to adjust the digital zoom amount for the image captured using the telephoto lens **12***c* with the lens magnification of 2.9 times, which is the selected lens.

This processing for adjusting the digital zoom amount will be described with reference to FIG. **11** and subsequent figures.

(Step S**9**)

FIG. **11** illustrates display data when the user slides the touched lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times to the right within the digital zoom adjustment section **107**.

The user slides (moves) the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times to a lens selection icon for a super-telephoto lens **12***d* with a lens magnification of 4.4 times.

According to this slide processing, the numerical value displayed on the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times is changed.

In (step S**9**) of FIG. **11**, the lens selection icon for the telephoto lens **12***c* with the lens magnification of 2.9 times touched by the user is slid to the right, and (×3.5) is displayed on the icon at this position.

This indicates that digital zoom processing for enlarging the image captured using the telephoto lens **12***c* with the lens magnification of 2.9 times to an image corresponding to an image captured using a lens with a lens magnification of 3.5 times is performed.

A digital zoom image of the image captured by the telephoto lens **12***c* with the lens magnification of 2.9 times touched by the user is displayed in the captured image display area **101**. That is, an image corresponding to an image captured using the lens with the lens magnification of 3.5 times is displayed.

The data processing unit of the imaging apparatus executes image correction processing for the image captured by the telephoto lens **12***c* with the lens magnification of 2.9 times depending on the slide amount of the lens selection icon slid by the user. The data processing unit of the imaging apparatus executes processing for correcting the captured image, that is, processing for extracting a partial area from the captured image and processing for enlarging an extracted image as digital zoom processing, and displays an image after the processing in the captured image display area **101**.

The image displayed in the captured image display area in FIG. **11** (step S9) is the image subjected to this digital zoom processing.

That is, the image is an image generated by executing digital zoom processing for extracting a partial area from the image captured by the telephoto lens **12***c* with the lens magnification of 2.9 times and enlarging the extracted image. This image corresponds to the image captured using the lens with the lens magnification of 3.5 times.

When the user confirms the image displayed in the captured image display area **101**, that is, the same digital zoom image as the image captured using the lens with the lens magnification of 3.5 times, and determines that the user desires to record this image, the user touches the shutter **103**.

(Step S**10**)

FIG. **12** (step S**10**) illustrates a state in which the user touches the shutter **103**. That is, FIG. **12** is a diagram illustrating a user operation when the user determines that the user desires to record the image displayed in the captured image display area **101**.

The image displayed in the captured image display area **101** of the display unit **11** of the imaging apparatus **10** in (step S**10**) illustrated in FIG. **12** is the image displayed in (step S**9**) described with reference to FIG. **11**.

That is, the image is a digital zoom processing image for an image captured by a telephoto lens with a lens magnification of 2.9 times. This image is the same as the image captured using the lens with the lens magnification of 3.5 times.

The user confirms this displayed image and touches the shutter **103** when the user determines that the user desires to record this image.

Through this touch processing, processing for capturing the image displayed in the captured image display area **101** is executed, and the image is recorded in the memory.

The image recorded in the memory is the digital zoom processing image for an image captured by a telephoto lens with a lens magnification of 2.9 times.

Thus, the imaging apparatus **10** of the present disclosure displays the lens selection portion (lens selection icon) **105** for selecting a lens to be used for photographing, in the operation portion display area **102** of the display unit **11**. Further, when the user touches one lens selection portion (lens selection icon) **105**, the digital zoom adjustment section **107** in which the selected one lens selection portion (lens selection icon) is slidable is displayed.

Thereafter, when the lens selection portion (lens selection icon) selected by the user is slid within the digital zoom adjustment section **107**, a value of the lens magnification corresponding to a virtual use lens realized by the digital zoom processing is displayed in the lens selection icon.

Further, an image (through image) captured using the selected lens corresponding to the lens selection portion (lens selection icon) **105** selected by the user is displayed in the captured image display area **101** of the display unit **11**. Further, when the lens selection portion (lens selection icon) selected by the user is slid within the digital zoom adjustment section **107**, an image subjected to digital zoom processing depending on the slide amount, that is, enlargement processing is displayed.

These processes allow the user to capture an ideal subject image of the user and record the subject image in the memory only through selection processing of the lens selection portion (lens selection icon) **105** and the slide processing.

That is, the user does not need to perform processing for switching between UIs of the operation portion when performing two processing including the lens selection processing and the digital zoom processing, can continuously execute the two processing through an operation with respect to the same icon, thereby eliminating a need for processing time of UI switching or the like, and perform photographing processing without missing a shutter chance.

An operation example of the lens selection portion (lens selection icon) **105** by the user and transition of display data of the operation portion according to a user operation will be collectively described with reference to FIGS. **13** to **16**.

FIG. **13** illustrates a section in which the wide-angle lens selection portion **105***b* is slidable.

When the user touches the wide-angle lens selection portion **105***b*, the digital zoom adjustment section **107** corresponding to the section in which the wide-angle lens selection portion **105***b* is slidable is displayed.

The digital zoom adjustment section **107** corresponding to the section in which the wide-angle lens selection portion **105***b* is slidable is a section from a wide-angle lens selection portion **105***b* (min) at a left end **p1** illustrated in FIG. **13** to a wide-angle lens selection portion **105***b* (max) at a right end **p2** illustrated in FIG. **13**, and the wide-angle lens selection portion **105***b* becomes slidable in this section (**p1** to **p2**).

When the wide-angle lens selection portion **105***b* is slid within this section (**p1** to **p2**) in which the wide-angle lens selection portion **105***b* is slidable, displayed numerical values of the wide-angle lens selection portion **105***b* are sequentially updated and displayed.

The displayed numerical value of the wide-angle lens selection portion **105***b* is a numerical value indicating a virtual lens magnification when a captured image or a digital zoom image is captured. The numerical values are sequentially updated and displayed between 0.7 and 1.0.

A position **p1** of "0.7" among the numerical values (0.7 to 1.0) displayed at the respective slide positions is a position of the wide-angle lens selection portion **105***b* (min) at the left end, and is a position at which a slide amount of the wide-angle lens selection portion **105***b* is zero.

The captured image itself of the wide-angle lens **12***b* with a lens magnification of 0.7 times is displayed in the captured image display area **101** at the position of the wide-angle lens selection portion **105***b* (min) at the left end.

That is, as shown in (b(min)) in a lower part of FIG. **13**, the captured image is an image itself captured by the wide-angle lens **12***b*, and the captured image not subjected to the digital zoom processing is displayed.

Meanwhile, the numerical value displayed in the wide-angle lens selection portion **105***b* becomes a numerical value greater than 0.7 and equal to or smaller than 1.0 at a position other than the wide-angle lens selection portion **105***b* (min) at the left end to which the wide-angle lens selection portion **105***b* is slid. A maximum lens magnification of 1.0 times is displayed at a position of the wide-angle lens selection portion **105***b* (max) at the right end **p2** illustrated in FIG. **13**.

The digital zoom processing image is displayed in the captured image display area **101** in a section in which the wide-angle lens selection portion **105***b* is slid. That is, a digital zoom image obtained by correcting the image captured by the wide-angle lens **12***b* with the lens magnification of 0.7 times is displayed. This digital zoom image is a corrected image obtained by cutting out and enlarging a part of the image captured by the wide-angle lens **12***b* with a lens magnification of 0.7 times.

When the wide-angle lens selection portion **105***b* is slid to the position **p2** of the wide-angle lens selection portion **105***b* (max) at the right end, the digital zoom processing image as shown in (b(max)) in a lower part of FIG. **13** is displayed in the captured image display area **101**.

That is, the corrected image generated by the digital zoom processing for the image captured by the wide-angle lens **12***b* with the lens magnification of 0.7 times is displayed. Specifically, a corrected digital zoom image that is the same as an image captured using a virtual lens with a lens magnification of 1.0 times is displayed.

FIG. **14** illustrates a section in which a standard lens selection portion **105***a* is slidable.

When the user touches the standard lens selection portion **105***a*, a digital zoom adjustment section **107** corresponding to the section in which the standard lens selection portion **105***a* is slidable is displayed.

The digital zoom adjustment section **107** corresponding to the section in which the standard lens selection portion **105***a* is slidable is a section from a standard lens selection portion **105***a* (min) at a left end **p3** illustrated in FIG. **14** to a standard lens selection portion **105***a* (max) at a right end **p4** illustrated in FIG. **14**, and the standard lens selection portion **105***a* becomes slidable in this section (**p3** to **p4**).

When the standard lens selection portion **105***a* is slid within this section (**p3** to **p4**) in which the standard lens selection portion **105***a* is slidable, displayed numerical values of the standard lens selection portion **105***a* are sequentially updated and displayed.

The numerical value displayed in the standard lens selection portion **105***a* is a numerical value indicating a virtual lens magnification when a captured image or a digital zoom image is captured. The numerical values are sequentially updated and displayed between 1.0 and 2.9.

A position **p3** of "1.0" among the numerical values (1.0 to 2.9) displayed at the respective slide positions is a position of the standard lens selection portion **105***a* (min) at the left end, and is a position at which a slide amount of the standard lens selection portion **105***a* is zero.

The captured image itself of the standard lens **12***a* with lens magnification of 1.0 times is displayed in the captured image display area **101** at the position of the standard lens selection portion **105***a* (min) at the left end.

That is, as shown in (a(min)) in a lower part of FIG. **14**, the captured image is an image itself captured by the standard lens **12***a*, and the captured image not subjected to the digital zoom processing is displayed.

On the other hand, the numerical value displayed in the standard lens selection portion **105***a* is a numerical value greater than 1.0 and equal to or smaller than 2.9 at a position other than the standard lens selection portion **105***a* (min) at the left end to which the standard lens selection portion **105***a* is slid. A maximum lens magnification of 2.9 times is displayed at a position of the standard lens selection portion **105***a* (max) at the right end **p4** illustrated in FIG. **14**.

The digital zoom processing image is displayed in the captured image display area **101** in a section in which the standard lens selection portion **105***a* is slid.

That is, a digital zoom image obtained by correcting the image captured by the standard lens **12***a* with the lens magnification of 1.0 times is displayed. This digital zoom image is a corrected image obtained by cutting out and enlarging a part of the image captured by the standard lens **12***a* with the lens magnification of 1.0 times.

When the standard lens selection portion **105***a* is slid to the position p**4** of the standard lens selection portion **105***a* (max) at the right end, the digital zoom processing image as shown in (a(max)) in the lower part of FIG. **14** is displayed in the captured image display area **101**.

That is, the corrected image generated by the digital zoom processing for the image captured by the standard lens **12***a* with the lens magnification of 1.0 times is displayed. Specifically, a corrected digital zoom image that is the same as an image captured using a virtual lens with a lens magnification of 2.9 times is displayed.

FIG. **15** illustrates a section in which the telephoto lens selection portion **105***c* is slidable.

When the user touches the telephoto lens selection portion **105***c*, the digital zoom adjustment section **107** corresponding to the section in which the telephoto lens selection portion **105***c* is slidable is displayed.

The digital zoom adjustment section **107** corresponding to the section in which the telephoto lens selection portion **105***c* is slidable is a section from a telephoto lens selection portion **105***c* (min) at the left end p**5** illustrated in FIG. **15** to a telephoto lens selection portion **105***c* (max) at a right end p**6** illustrated in FIG. **15**, and the telephoto lens selection portion **105***c* becomes slidable in this section (p**5** to p**6**).

When the telephoto lens selection portion **105***c* is slid within this section (p**5** to p**6**) in which telephoto lens selection portion **105***c* is slidable, displayed numerical values of the telephoto lens selection portion **105***c* are sequentially updated and displayed.

The displayed numerical value of the telephoto lens selection portion **105***c* is a numerical value indicating a virtual lens magnification when a captured image or a digital zoom image is captured. The numerical values are sequentially updated and displayed between 2.9 and 4.4.

A position p**5** of "2.9" among numerical values (2.9 to 4.4) displayed at the respective slide positions is a position of the telephoto lens selection portion **105***c* (min) at the left end, and is a position at which a slide amount of the telephoto lens selection portion **105***c* is 0.

The captured image itself of the telephoto lens **12***c* with the lens magnification of 2.9 times is displayed in the captured image display area **101** at the position of the telephoto lens selection portion **105***c* (min) at the left end.

That is, as shown in (c(min)) in a lower part of FIG. **15**, the image is the image itself captured by the telephoto lens **12***c*, and the captured image not subjected to the digital zoom processing is displayed.

Meanwhile, the numerical value displayed in the telephoto lens selection portion **105***c* is a numerical value greater than 2.9 and equal to or smaller than 4.4 at a position other than the telephoto lens selection portion **105***c* (min) at the left end to which the telephoto lens selection portion **105***c* is slid. A maximum lens magnification of 4.4 is displayed at a position of the telephoto lens selection portion **105***c* (max) at the right end p**6** illustrated in FIG. **15**.

The digital zoom processing image is displayed in the captured image display area **101** in a section in which the telephoto lens selection portion **105***c* is slid.

That is, a digital zoom image obtained by correcting the image captured by the telephoto lens **12***c* with the lens magnification of 2.9 times is displayed. This digital zoom image is a corrected image obtained by cutting out and enlarging a part of the image captured by the telephoto lens **12***c* with the lens magnification of 2.9 times.

When the telephoto lens selection portion **105***c* is slid to the position p**6** of the telephoto lens selection portion **105***c* (max) at the right end, the digital zoom processing image as shown in (c(max)) in the lower part of FIG. **15** is displayed in the captured image display area **101**.

That is, the corrected image generated by the digital zoom processing for the image captured by the telephoto lens **12***c* with the lens magnification of 2.9 times is displayed. Specifically, a corrected digital zoom image that is the same as an image captured using a virtual lens with a lens magnification of 4.4 times is displayed.

FIG. **16** illustrates a section in which the super-telephoto lens selection portion **105***d* is slidable.

When the user touches the super-telephoto lens selection portion **105***d*, the digital zoom adjustment section **107** corresponding to the section in which the super-telephoto lens selection portion **105***d* is slidable is displayed.

The digital zoom adjustment section **107** corresponding to the section in which the super-telephoto lens selection portion **105***d* is slidable is a section from a super-telephoto lens selection portion **105***d* (min) at a left end p**7** illustrated in FIG. **16** to a section of the super-telephoto lens selection portion **105***d* (max) at a right end p**8** illustrated in FIG. **16**, and the super-telephoto lens selection portion **105***d* is slidable in this section (p**7** to p**8**).

When the super-telephoto lens selection portion **105***d* is slid within this section (p**7** to p**8**) in which the super-telephoto lens selection portion **105***d* is slidable, the displayed numerical values of the super-telephoto lens selection portion **105***d* are sequentially updated and displayed.

The displayed numerical value of the super-telephoto lens selection portion **105***d* is a numerical value indicating a virtual lens magnification when a captured image or a digital zoom image is captured. The numerical values are sequentially updated and displayed between 4.4 and 8.3.

A position p**7** of "4.4" among the numerical values (4.4 to 8.3) displayed at the respective slide positions is a position of the super-telephoto lens selection portion **105***d* (min) at the left end, and is a position at which a slide amount of the super-telephoto lens selection portion **105***d* is zero.

The image itself captured by the super-telephoto lens **12***d* with a lens magnification of 4.4 is displayed in the captured image display area **101** at the position of the super-telephoto lens selection portion **105***d* (min) at the left end.

That is, as shown in (d(min)) in a lower part of FIG. **16**, the captured image is an image itself captured by the super-telephoto lens **12***d*, and the captured image not subjected to the digital zoom processing is displayed.

On the other hand, a numerical value displayed in the super-telephoto lens selection portion **105***d* becomes a numerical value greater than 4.4 and equal to or smaller than 8.3 at a position other than the super-telephoto lens selection portion **105***d* (min) at the left end to which the super-telephoto lens selection portion **105***d* is slid. A maximum lens magnification of 8.3 times is displayed at a position of the super-telephoto lens selection portion **105***d* (max) at the right end p**8** illustrated in FIG. **16**.

The digital zoom processing image is displayed in the captured image display area **101** in a section in which the super-telephoto lens selection portion **105***d* is slid. That is, a digital zoom image obtained by correcting the image captured by the super-telephoto lens **12***d* with a lens magnification of 4.4 is displayed. This digital zoom image is a corrected image obtained by cutting out and enlarging a part of the image captured by the super-telephoto lens **12***d* with a lens magnification of 4.4.

When the super-telephoto lens selection portion **105***d* is slid to the position p**8** of the super-telephoto lens selection portion **105***d* (max) at the right end, the digital zoom processing image as shown in (d(max)) in the lower part of FIG. **16** is displayed in the captured image display area **101**.

That is, the corrected image generated by the digital zoom processing for the image captured by the super-telephoto lens **12***d* with a lens magnification of 4.4 times is displayed. Specifically, a corrected digital zoom image that is the same as an image captured using a virtual lens with a lens magnification of 8.3 times is displayed.

## 3. OPERATION PORTION UI DISPLAY CONTROL PROCESSING SEQUENCE EXECUTED BY IMAGING APPARATUS OF PRESENT DISCLOSURE

Next, an operation portion UI display control processing sequence executed by the imaging apparatus of the present disclosure will be described.

The operation portion UI display control processing sequence executed by the imaging apparatus of the present disclosure will be described with reference to a flowchart illustrated in FIG. **17**.

Processing according to a flow illustrated in FIG. **17** can be executed in the data processing unit of the imaging apparatus **10** according to a program stored in the storage unit of the imaging apparatus **10**. The data processing unit includes, for example, a processor such as a CPU having a program execution function, and can perform processing according to a flow through program execution processing using the processor.

Hereinafter, processing of each step shown in the flowchart of FIG. **17** will be described.

(Step S**101**)

First, in step S**101**, the data processing unit of the imaging apparatus **10** determines whether or not touch processing with respect to the lens selection portion (lens selection icon) **105** displayed in the operation portion display area **102** of the display unit **11** of the imaging apparatus **10** by the user has been detected.

When a determination is made that user touch processing with respect to the lens selection portion (lens selection icon) **105** has been detected, the processing proceeds to step S**102**.

On the other hand, when the user touch processing with respect to the lens selection portion (lens selection icon) **105** has not been detected in step S**101**, for example, when it has been detected that the user has taken away the finger from the lens selection portion **105**, the processing proceeds to step S**107** to determine whether or not the operation (touch) with respect to the shutter **103** by the user has been detected.

When the shutter operation (touch) by the user has not been detected, the processing returns to step S**101**.

On the other hand, when the shutter operation (touch) by the user has been detected, the processing proceeds to step S**108** to execute image capturing and record a captured image in the memory.

(Step S**102**)

When a determination is made in step S**101** that the touch processing with respect to the lens selection portion (lens selection icon) **105** by the user has been detected, the data processing unit of the imaging apparatus **10** executes processing for switching an image capturing lens to the lens touched by the user, that is, the lens selected by the user in step S**102**.

For example, when the lens selection portion (lens selection icon) touched by the user is an icon for a telephoto lens (lens magnification of 2.9 times), processing for switching the photographing lens to the telephoto lens is performed.

This state corresponds to, for example, the state (step S**7**) in FIG. **10** described above.

(Step S**103**)

Next, in step S**103**, the data processing unit of the imaging apparatus **10** displays the digital zoom ON state identification icon **106** on the display unit **11** of the imaging apparatus **10**, and displays the digital zoom adjustment section **107** in which the touched lens selection portion is slidable.

This state corresponds to, for example, the state (step S**8**) in FIG. **10** described above.

(Step S**104**)

Next, in step S**104**, the data processing unit of the imaging apparatus **10** detects whether or not the touched lens selection portion has been slid by the user.

When it has been detected that the lens selection portion has been slid by the user, the processing proceeds to step S**105**.

On the other hand, when the lens selection portion has not been slid by the user, the processing returns to step S**101** to execute the following processing.

Processing (a) to (c2) to be described below are processing that are executed when a determination is made in step S**104** that the lens selection portion has not been slid by the user.

(a) The processing returns to step S**101** to determine whether or not a new touch with respect to the lens selection portion by the user has been detected.

(b1) When a determination is made in step S**101** that the new touch with respect to the lens selection portion by the user has been detected, the processing proceeds to step S**102**.

(b2) On the other hand, when new user touch processing with respect to the lens selection portion (lens selection icon) **105** has not been detected in step S**101**, for example, when it has been detected that the user has taken away the finger from the lens selection portion **105**, the processing proceeds to step S**107** to determine whether or not the operation (touch) with respect to the shutter **103** by the user has been detected.

(c1) When the shutter operation (touch) by the user has not been detected, the processing returns to step S**101**.

(c2) On the other hand, when the shutter operation (touch) by the user has been detected, the processing proceeds to step S**108** to execute image capturing and record a captured image in the memory.

Thus, when a determination is made in step S**104** that the touched lens selection portion is not slid by the user, the processing (a) to (c2) are executed.

On the other hand, when it has been detected in step S**104** that the lens selection portion has been slid by the user, the processing proceeds to step S**105**. Hereinafter, processing of step S**105** and subsequent steps will be described.

(Steps S**105** and S**106**)

In step S**104**, when the slide processing for the lens selection portion (lens selection icon) **105** by the user has been detected, the data processing unit of the imaging apparatus **10** executes the following processing in steps S**105** and S**106**.

In step S**105**, the data processing unit of the imaging apparatus **10** executes digital zoom processing (enlargement processing) for the captured image depending on the slide amount of the lens selection portion (lens selection icon) of the user, and displays the digital zoom processing image in the captured image display area **101** of the display unit **11**.

Further, in step S**106**, the data processing unit of the imaging apparatus **10** executes processing for updating the lens magnification displayed in the lens selection portion (lens selection icon) slid by the user to a numerical value depending on the slide amount (digital zoom processing amount).

A state in which steps S**105** and S**106** are executed corresponds to, for example, the state of (step S9) in FIG. **11** described above.

The processing of step S**105** and the processing of step S**106** may be executed in parallel.

After the processing of steps S**105** and S**106**, the processing returns to step S**101** to determine whether or not the new touch with respect to the lens selection portion by the user has been detected.

When a determination is made that the new touch with respect to the lens selection portion by the user has been detected, the processing proceeds to step S**102**.

On the other hand, when the new user touch processing with respect to the lens selection portion (lens selection icon) **105** has not been detected, the processing proceeds to step S**107**. For example, when it has been detected that a finger of the user has taken away the lens selection portion (lens selection icon) **105**, the processing proceeds to step S**107**.

(Step S**107**)

The processing in step S**107** is processing that is executed when the user touch processing with respect to the lens selection portion (lens selection icon) **105** has not been detected in step S**101**, for example, when it has been detected that the finger of the user has taken away the lens selection portion **105**.

In this case, the data processing unit of the imaging apparatus **10** determines whether or not touch of the shutter **103** by the user has been detected in step S**107**.

When the shutter operation (touch) by the user has not been detected, the processing returns to step S**101**.

On the other hand, when the shutter operation (touch) by the user has been detected, the processing proceeds to step S**108**.

(Step S**108**)

When the shutter operation (touch) by the user has been detected in step S**107**, the data processing unit of the imaging apparatus **10** executes image capturing and records the captured image in the memory in step S**108**.

This image capturing processing corresponds to, for example, the image capturing processing of (step S**6**) in FIG. **9** or the image capturing processing of (step S**10**) in FIG. **12** described above.

As described above, the imaging apparatus **10** of the present disclosure displays the lens selection portion (lens selection icon) **105** for selecting a lens to be used for photographing, in the operation portion display area **102** of the display unit **11**, and displays the digital zoom adjustment section **107** in which the selected one lens selection portion (lens selection icon) is slidable when the user touches one lens selection portion (lens selection icon) **105**.

Further, when the lens selection portion (lens selection icon) selected by the user is slid within the digital zoom adjustment section **107**, a value of the lens magnification corresponding to a virtual use lens realized by the digital zoom processing is displayed in the lens selection icon.

Further, the image (through image) captured using the selected lens corresponding to the lens selection portion (lens selection icon) **105** selected by the user is displayed in the captured image display area **101** of the display unit **11**. Further, when the lens selection portion (lens selection icon) selected by the user is slid within the digital zoom adjustment section **107**, an image subjected to digital zoom processing depending on the slide amount is displayed.

These processes allow the user to capture an ideal subject image of the user and record the subject image in the memory only through selection processing of the lens selection portion (lens selection icon) **105** and the slide processing.

## 4. DISPLAY CONTROL PROCESSING FOR NOTIFYING USER OF IMAGE QUALITY DEGRADATION DUE TO DIGITAL ZOOM

Next, display control processing for notifying the user of the deterioration of the image quality due to digital zoom executed by the imaging apparatus **10** of the present disclosure will be described.

As described above, the digital zoom processing is performed as the image enlargement processing for a captured image. For example, a portion of the image captured using the standard lens is cropped and enlarged so that it is possible to generate an intermediate image between the image captured using the standard lens and the image captured using the telephoto lens.

A specific example of image enlargement processing in the digital zoom processing will be described with reference to FIG. **18**.

In FIG. **18**, a captured image (min) captured with the standard lens selection portion **105***a* set to a position (lens magnification=1.0) at a left end **p3** in a slide section, and a captured image (max) at a position (lens magnification of 2.9 times) at a right end **p4** in the slide section of the standard lens selection portion **105***a* are shown.

The captured image (min) captured with the standard lens selection portion **105***a* set to the position (lens magnification=1.0) at the left end p**3** of the slide section is "captured image (min)" in a lower part. This is the image itself captured by the standard lens **12***a*, and is a captured image not subjected to the digital zoom processing.

On the other hand, the captured image (max) captured with the standard lens selection portion **105***a* set to the position (lens magnification of 2.9 times) at the right end p**4** of the slide section is a "captured image (max)" in the lower part. This is a digital zoom processing image of the image captured by the standard lens **12***a*. That is, a dotted-line rectangular area of the "captured image (min)" shown on the left side in the lower part of FIG. **18** is cut out and enlarged so that the "captured image (max)" shown on the right side in the lower part of FIG. **18** is generated.

Thus, the digital zoom processing image can be generated by enlarging a partial area of the original image, that is, the captured image not subjected to the digital zoom processing, at an enlargement ratio according to the digital zoom amount.

When the number of pixels of a captured image not subjected to enlargement processing at all, for example, the "captured image (min)" is "n×m", the number of pixels of a dotted line image area extracted in the digital zoom processing is the number of pixels smaller than "n×m", for example "s×t".

That is, since the digital zoom image generated with the enlargement processing is an image of "n×m" pixels obtained by interpolating data of an s×t pixel area cut out from an "n×m" pixel area input from an image sensor by the digital zoom processing with data from neighboring pixels, the image quality or resolution is impaired as compared with an image stored without being subjected to digital zoom from original "n×m" pixels.

A bottom right of FIG. **19** shows the following two images.

(Image A) An image captured with the standard lens selection portion **105***a* set to the position (lens magnification of 2.9 times) at the right end p**4** of the slide section

(Image B) An image captured with the telephoto lens selection portion **105***c* not slid

(Image A) is a digital zoom processing image. That is, (image A) is a digital zoom processing image generated by cutting out and enlarging a dotted-line rectangular area of the "captured image (min)" captured with (lens magnification=1.0) without the standard lens selection portion **105***a* shown at the lower left of FIG. **19** being slid.

On the other hand, (Image B) is the image captured with the telephoto lens selection portion **105***c* not slid, and is an image not subjected to the digital zoom processing.

Both (Image A) and (Image B) correspond to images captured using the lens with the lens magnification of 2.9 times, and are images having the same setting of an angle of view of the image or the same size of a subject.

However, (image A) is a digital zoom image obtained by cutting out and enlarging a partial area (s×t pixels) of the original captured image (n×m pixels), and is an image generated on the basis of pixel data of s×t pixels smaller than n×m pixels. On the other hand, (image B) is not subjected to the digital zoom processing (enlargement processing), and has a data amount corresponding to the number of constituent pixels (n×m pixels) of the original captured image.

That is, (image A) subjected to the digital zoom processing (enlargement processing) is an image generated on the basis of less pixel data than (image B) not subjected to the digital zoom processing (enlargement processing).

As a result, (image A) subjected to the digital zoom processing (enlargement processing) is an image whose image quality is lower than (image B) not subjected to the digital zoom processing.

However, the user may capture the digital zoom image without noticing such deterioration of the image quality in many cases.

Especially, it is difficult to clearly recognize the deterioration of the image quality on a small screen of a smartphone or the like, the digital zoom tends to be used frequently.

An embodiment to be described below is an embodiment in which display control for notifying the user of the deterioration of the image quality due to the digital zoom processing is performed.

The imaging apparatus **10** of the present disclosure executes display processing for notifying the user of the deterioration of the image quality due to the digital zoom processing in the operation portion display area **102** of the display unit **11**. Hereinafter, an example of a plurality of pieces of display data for notification of the deterioration of the image quality will be described.

FIGS. **20**A, **20**B, and **20**C are diagrams illustrating an example of display data for the notification of deterioration of image quality due to the digital zoom processing.

In FIGS. **20**A, **20**B, and **20**C, states in which the user slides the standard lens selection portion **105***a* from a position at a left end (lens magnification=1.0) to the right are shown as FIGS. **20**A, **20**B, and **20**C.

FIG. **20**A is a state in which the standard lens selection portion **105***a* has been set to the left end position (lens magnification=1.0) of the slide section. That is, the slide amount is 0, and an image captured at this position is an image not subjected to the digital zoom processing.

The data processing unit of the imaging apparatus **10** of the present disclosure sets an output light of the standard lens selection portion **105***a* to white when the standard lens selection portion **105***a* is at this position (a left end of the slide section).

The output light of the standard lens selection portion **105***a* is set to white so that the user is notified that the digital zoom amount is 0 and image quality is not degraded.

FIG. **20**B is a state in which the standard lens selection portion **105***a* is slid from the left end position (lens magnification=1.0) to the right and moved to approximately a center of the slide section. That is, the slide amount is approximately (½) of the slide section. An image captured at this position is an image subjected to the digital zoom processing, that is, image cut-out and enlargement processing. However, the enlargement ratio is low and the level of the deterioration of the image quality is low.

The data processing unit of the imaging apparatus **10** of the present disclosure sets the output light of the standard lens selection portion **105***a* to yellow when the standard lens selection portion **105***a* is at this position (approximately at the center of the slide section).

The output light of the standard lens selection portion **105***a* is set to yellow so that the user is notified that the digital zoom amount is small, but the image quality is somewhat degraded.

FIG. **20**C is a state in which the standard lens selection portion **105***a* has been further slid to the right and moved approximately to a right end of the slide section. That is, the slide amount is the substantially entire slide section. The image captured at this position is the image subjected to the digital zoom processing, that is, image cut-out and enlargement processing, and has a high enlargement ratio and a high level of the deterioration of the image quality.

The data processing unit of the imaging apparatus **10** of the present disclosure sets the output light of the standard lens selection portion **105***a* to red when the standard lens selection portion **105***a* is at this position (approximately at the right end of the slide section).

The output light of the standard lens selection portion **105***a* is set to red so that the user is notified that the digital zoom amount is large and the image quality is greatly degraded.

Thus, the data processing unit of the imaging apparatus **10** of the present disclosure executes display control for changing the output light of the lens selection portion **105** depending on the slide amount of the lens selection portion **105**. Such display control is performed so that it becomes possible to notify the user of the digital zoom amount or the deterioration of the image quality level depending on the slide amount of the lens selection portion **105**.

With this processing, the user can take measures such as switching the lens to be used to another lens when the deterioration of the image quality is large.

FIGS. **21**A, **21**B, and **21**C are diagrams illustrating an example of different display data for notifying the deterioration of the image quality due to the digital zoom processing.

In FIGS. **21**A, **21**B, and **21**C, states in which the user slides the standard lens selection portion **105***a* from a position at a left end (lens magnification=1.0) to the right are shown as FIGS. **21**A, **21**B, and **21**C, similar to FIGS. **20**A, **20**B, and **20**C.

FIG. **21**A is a state in which the standard lens selection portion **105***a* has been set to the left end position (lens magnification=1.0) of the slide section.

FIG. **21**B is a state in which the standard lens selection portion **105***a* is slid from the left end position (lens magnification=1.0) to the right and moved to approximately the center of the slide section.

FIG. **21**C is a state in which the standard lens selection portion **105***a* has been further slid to the right and moved approximately to the right end of the slide section. These are the three states.

In FIG. **21**A, the data processing unit of the imaging apparatus **10** of the present disclosure sets the output light of the standard lens selection portion **105***a* to white.

The output light of the standard lens selection portion **105***a* is set to white so that the user is notified that the digital zoom amount is 0 and that image quality is not degraded.

This display control processing is the same display control processing as in FIG. **20**A.

In the state FIG. **21**B, the output light of the standard lens selection portion **105***a* is set to yellow, and the shape is also changed. That is, a shape of the standard lens selection portion **105***a* is set to a rounded square shape.

In the state FIG. **21**C, the output light from the standard lens selection portion **105***a* is set to red, and the shape thereof is changed to a square shape.

The example illustrated in FIGS. **21**A, **21**B, and **21**C shows a configuration that performs display control processing for changing the shape of the standard lens selection portion **105***a*, in addition to the change in the color of the output light of the standard lens selection portion **105***a*.

Such display control is performed so that it becomes possible to notify the user of the digital zoom amount or the deterioration of the image quality level depending on the slide amount of the lens selection portion **105** in an easy-to-understand manner.

Further, as illustrated in FIGS. **22**A, **22**B, and **22**C, a configuration that performs display control for changing the size of the standard lens selection portion **105***a*, in addition to the change in the color and shape of the output light of the standard lens selection portion **105***a*.

In FIGS. **22**A, **22**B, and **22**C, the following three states are also illustrated.

FIG. **22**A is a state in which the standard lens selection portion **105***a* has been set to the left end position (lens magnification=1.0) of the slide section.

FIG. **22**B is a state in which the standard lens selection portion **105***a* is slid from the left end position (lens magnification=1.0) to the right and moved to approximately a center of the slide section.

FIG. **22**C is a state in which the standard lens selection portion **105***a* has been further slid to the right and moved approximately to the right end of the slide section.

In the state FIG. **22**A, the output light of the standard lens selection portion **105***a* is set to white.

The output light of the standard lens selection portion **105***a* is set to white so that the user is notified that the digital zoom amount is 0 and image quality is not degraded.

This display control processing is the same display control processing as in FIG. **20**A.

In the state FIG. **22**B, the output light of the standard lens selection portion **105***a* is set to yellow, the shape thereof is set to the rounded square shape, and a size of the standard lens selection portion **105***a* is increased.

In the state FIG. **22**C, the output light of the standard lens selection portion **105***a* is set to red, the shape thereof is changed to a square shape, and the size of the standard lens selection portion **105***a* is made larger than in the state FIG. **22**B.

The example illustrated in FIGS. **22**A, **22**B, and **22**C shows a configuration that performs display control processing for changing the size of the standard lens selection portion **105***a*, in addition to the change in the color and shape of the output light of the standard lens selection portion **105***a*.

Such display control is performed so that it becomes possible to notify the user of the digital zoom amount or the deterioration of the image quality level depending on the slide amount of the lens selection portion **105** in an easy-to-understand manner.

Further, a configuration that performs a notification using vibration as illustrated in FIGS. **23**A, **23**B, and **23**C may be adopted.

In FIGS. **23**A, **23**B, and **23**C, the following three states are also illustrated.

FIG. **23**A is a state in which the standard lens selection portion **105***a* has been set to the left end position (lens magnification=1.0) of the slide section.

FIG. **23**B is a state in which the standard lens selection portion **105***a* is slid from the left end position (lens magnification=1.0) to the right and moved to approximately a center of the slide section.

FIG. **23**C is a state in which the standard lens selection portion **105***a* has been further slid to the right and moved approximately to the right end of the slide section.

In the state FIG. **23**A, vibration processing is not performed.

A vibration is not generated so that the user is notified that the digital zoom amount is 0 and image quality is not degraded.

In the state FIG. **23**B, weak vibration is generated.

In the state FIG. **23**C, strong vibration is generated.

Thus, the user is notified of the digital zoom amount or the deterioration of the image quality level depending on the slide amount of the lens selection portion **105** by control of the presence or absence and intensity of the vibration.

Further, a configuration in which a melody is output and a notification is performed as illustrated in FIGS. **24**A, **24**B, and **24**C may be adopted.

In FIGS. **24**A, **24**B, and **24**C, the following three states are also illustrated.

FIG. **24**A is a state in which the standard lens selection portion **105***a* has been set to the left end position (lens magnification=1.0) of the slide section.

FIG. **24**B is a state in which the standard lens selection portion **105***a* is slid from the left end position (lens magnification=1.0) to the right and moved to approximately a center of the slide section.

FIG. **24**C is a state in which the standard lens selection portion **105***a* has been further slid to the right and moved approximately to the right end of the slide section.

In the state FIG. **24**A, melody output processing is not performed.

The melody output is not performed so that the user is notified that the digital zoom amount is 0 and the image quality is not degraded.

In the state FIG. **24**B, a weaker melody is output.

In the state FIG. **24**C, a strong melody is output.

Thus, the user is notified of the digital zoom amount or the deterioration of the image quality level depending on the slide amount of the lens selection portion **105** by control of the presence or absence of the melody output and the intensity of an output sound.

## 5. OTHER EXAMPLES OF DISPLAY CONTROL PROCESSING

Next, other display control processing examples different from the above-described embodiment will be described.

Four display control processing examples shown below will be sequentially described.

(a) An example of processing for displaying an icon for digital zoom adjustment independent from the lens selection portion so that the icon is available

(b) An example of processing for outputting a message indicating that the digital zoom is in an ON state

(c) An example of processing for displaying the slide section independent of the lens selection portion so that the slide section is available

(d) An example of processing for displaying a zoom adjustment section in an apparatus having a configuration capable of optical zooming

((a) an Example of Processing for Displaying an Icon for Digital Zoom Adjustment Independent from the Lens Selection Portion so that the Icon is Available)

First, "(a) an example of processing for displaying an icon for digital zoom adjustment independent from the lens selection portion so that the icon is available" will be described with reference to FIG. **25**.

As illustrated in FIG. **25**, the lens selection portion **105** displays the slidable digital zoom adjustment section **107**, and displays a zoom amount down instruction icon (W) **121** for decreasing the digital zoom amount and a zoom amount up instruction icon (T) **122** for increasing the digital zoom amount on the left and right of the digital zoom adjustment section **107**. "W" denotes "wide" and "T" denotes "tele".

The user can also adjust the digital zoom amount by sliding the lens selection portion **105** within the digital zoom adjustment section **107**, but the user can adjust the digital zoom amount by touching the zoom amount down instruction icon (W) **121** or the zoom amount up instruction icon (T) **122**.

When the user touches the zoom amount down instruction icon (W) **121**, the lens selection portion **105** can be moved to the left within the digital zoom adjustment section **107** so that the digital zoom amount can be decreased.

Further, when the user touches zoom amount up instruction icon (T) **122**, the lens selection portion **105** can be moved to the right within the digital zoom adjustment section **107** so that the digital zoom amount can be increased.

((b) an Example of Processing for Outputting a Message Indicating that the Digital Zoom is in an ON State)

Next, "(b) an example of processing for outputting a message indicating that the digital zoom is in an ON state" will be described with reference to FIG. **26**.

When the data processing unit of the imaging apparatus **10** detects a user touch with respect to the lens selection portion (lens selection icon) **105**, the data processing unit of the imaging apparatus **10** displays the digital zoom ON state identification icon **106** and displays the digital zoom adjustment section **107**, as illustrated in FIG. **26**.

Further, a digital zoom ON state notification message **125** is displayed.

The digital zoom ON state notification message **125** is, for example, a message such as

"Digital zoom is an ON state".

The user can reliably confirm a digital zoom processing situation from this message output.

((c) an Example of Processing for Displaying the Slide Section Independent of the Lens Selection Portion so that the Slide Section is Available)

Next, "(c) an example of processing for displaying the slide section independent of the lens selection portion so that the slide section is available" will be described with reference to FIG. **27**.

When the data processing unit of the imaging apparatus **10** detects a user touch with respect to the lens selection portion (lens selection icon) **105**, the data processing unit of the imaging apparatus **10** displays the digital zoom ON state identification icon **106** and displays the digital zoom adjustment section **107**, as illustrated in FIG. **27**.

However, in the present processing example, the lens selection portion (lens selection icon) **105** touched by the user and the digital zoom adjustment section **107** are displayed at positions different from the other lens selection portions (lens selection icons) **105**, as illustrated in FIG. **27**.

Such display processing enables the user to more clearly distinguish between and confirm a selected lens and a non-selected lens.

((d) an Example of Processing for Displaying a Zoom Adjustment Section in an Apparatus Having a Configuration Capable of Optical Zooming)

Next, an example of processing for displaying a zoom adjustment section in an apparatus having a configuration capable of optical zooming will be described.

In the above-described embodiments, the apparatus such as a smartphone having no zoom lens have been described. As described above, the apparatus such as a smartphone having no zoom lens performs digital zoom processing, for example, when the intermediate image between the image captured using the standard lens and the image captured using the telephoto lens is generated.

However, it becomes possible to achieve optical zooming with a configuration for bending an optical axis using a prism or the like even in a thin apparatus such as a smartphone. Specifically, it becomes possible to perform optical zooming even in the thin apparatus such as a smartphone by mounting a periscope type lens system (refractive optical system) having the configuration for bending an optical axis using a prism or the like.

For example, a configuration in which only one or two of the four types of lenses described in the above-described embodiment, that is, the wide-angle lens, the standard lens, the telephoto lens, and the super-telephoto lens are periscope type optical zoom lenses is assumed.

In the above-described embodiment, when the telephoto lens **12***c* with the lens magnification of 2.9 times is a periscope type optical zoom lens, it is possible to capture a clear image by the optical zooming without performing the digital zoom in an entire section corresponding to the lens magnification of 2.9 times to 4.4 times.

When a configuration capable of performing such optical zoom processing is adopted, it is preferable to perform a display for notifying the user that image capturing by the optical zooming is executed.

A display configuration that enables such notification will be described with reference to FIGS. **28**A and **28**B.

FIGS. **28**A and **28**B illustrates display examples of the following two sections in which slidable is possible.

FIG. **28**A Section in which the standard lens selection portion **105***a* is slidable

FIG. **28**B Section in which the telephoto lens selection portion **105***c* is slidable

It is assumed that the standard lens **12***a* is not a periscope type optical zoom lens, but the telephoto lens **12***c* is a periscope type optical zoom lens and is capable of optical zooming.

In this case, the section (p**3** to p**4**) in which the standard lens selection portion **105***a* is slidable, which is illustrated in FIG. **28**A, is displayed as a digital zoom adjustment section in which a digital zoom image corresponding to a lens magnification of 1.0 times to 2.9 as described with reference to FIG. **14** can be captured.

On the other hand, an "optical zoomable section identification data **151**" as illustrated in FIG. **28**B is additionally displayed in a section in which the telephoto lens selection portion **105***c* corresponding to the telephoto lens **12***c*, which is a periscope type optical zoom lens and is capable of optical zooming, is slidable.

The user can view this "optical zoomable section identification data **151**" and recognize that the clear image by the optical zooming is captured.

Further, an optical zoom ON state identification icon **152** may be displayed on the display unit **11** as illustrated in FIG. **29**. The user can confirm the optical zoom ON state identification icon **152** to recognize that the clear image by the optical zooming is captured.

Although the example described with reference to FIGS. **28**A, **28**B, and **29** is a setting example in which the telephoto lens **12***c* is a lens capable of optical zooming, and the standard lens **12***a* is a lens incapable of optical zooming, the lens capable of optical zooming may be all or part of the lens.

For example, a configuration in which all lenses are optical zoomable lenses is also possible. In this case, the optical zoomable section identification data **151** as illustrated in FIG. **28**B is displayed in an entire section in which the lens selection portion is slidable. Further, the optical zoom ON state identification icon **152** illustrated in FIG. **29** is also displayed.

Further, there is a configuration in which optical zooming is possible only at a point of a specific lens magnification even with a periscope type optical zoom lens. In the case of such a configuration, it is necessary to notify the user of a position at which the optical zooming is possible.

FIG. **30** is a diagram illustrating an example of display data capable of notifying the user of a position at which optical zooming is possible.

As illustrated in FIG. **30**, a plurality of "optical zoomable section identification pieces of data **153**" are displayed discretely in the section in which the telephoto lens selection portion **105***c* corresponding to the telephoto lens **12***c* capable of optical zooming is slidable only at a specific lens magnification point.

The user can view this "optical zoomable section identification data **153**" and recognize that the clear image by the optical zooming is captured only at the plurality of positions.

Further, in this case, it is preferable to perform a setting so that either the digital zoom ON identification icon **106** or the optical zoom ON state identification icon **152** is displayed on the display unit **11** as illustrated in FIGS. **31**A and **31**B depending on a slide position of the telephoto lens selection portion **105***c*.

When the slide position of the telephoto lens selection portion **105***c* in the section in which the telephoto lens selection portion **105***c* is slidable is at an optical zoomable position, the optical zoom ON state identification icon **152** is displayed as illustrated in FIG. **31**B.

On the other hand, when the slide position of the telephoto lens selection portion **105***c* is a position at which the optical zooming is not possible, the digital zoom ON state identification icon **106** is displayed as illustrated in FIG. **31**A.

The user can reliably determine whether the zoom position is a zoom position at which image capturing by the optical zooming is executed or a zoom position at which an image capturing by the digital zoom is executed, depending on such display data.

Further, a case in which a display mode of the icon of the lens selection portion is changed when the lens selection portion is slid to the position at which optical zooming is possible may be adopted.

A specific example is illustrated in FIGS. **32**A and **32**B. FIGS. **32**A and **32**B illustrate the following two diagrams.

FIG. **32**A Display data when the slide position of the telephoto lens selection portion **105***c* is a digital zoom position

FIG. **32**B Display data when the slide position of the telephoto lens selection portion **105***c* is an optical zoom position

The telephoto lens selection portion display data in the case of FIG. **32**A, that is, a case in which the slide position of the telephoto lens selection portion **105***c* is the digital zoom position is display data for the normal telephoto lens selection portion **105***c*.

On the other hand, the telephoto lens selection portion display data in the case of FIG. **32**B, that is, a case in which the slide position of the telephoto lens selection portion **105***c* is the optical zoom position is display data different from the normal telephoto lens selection portion **105***c*.

That is, the display data is displayed as a telephoto lens selection portion with an optical zoomable position ancestor identification display **155**.

Thus, a configuration in which a display mode of the lens selection portion is changed when the lens selection portion is slid to the position at which the optical zooming is possible may be adopted.

## 6. CONFIGURATION EXAMPLE OF IMAGING APPARATUS OF PRESENT DISCLOSURE

Next, a configuration example of the imaging apparatus of the present disclosure will be described.

The configuration example of the imaging apparatus of the present disclosure will be described with reference to FIG. **33**.

An imaging apparatus **200** illustrated in FIG. **33** corresponds to the imaging apparatus **10** described in the above embodiment.

As illustrated in FIG. **33**, the imaging apparatus **200** includes an operation information analysis unit **201**, a lens switching control unit **202**, an imaging unit **203**, an image processing unit **204**, a memory (storage unit) **205**, a display control unit **206**, a display unit **210**, and an apparatus control unit **215**.

The display control unit **206** includes a captured image display control unit **207** and an operation portion (UI portion) display control unit **208**.

Further, the display unit **210** includes an image display unit **211** and an operation portion (UI portion) display unit **212**.

The operation information analysis unit **201** analyzes a user operation with respect to the operation portion (UI portion) display unit **212** displayed on the display unit **210**.

For example, the operation information analysis unit **201** detects and analyzes the touch processing or the slide processing for the lens selection portion (lens selection icon)

**105**, and the touch processing with respect to the shutter **103** described in the previous embodiment.

User operation information analyzed by the operation information analysis unit **201** is output to the lens switching control unit **202**, the imaging unit **203**, the image processing unit **204**, the captured image display control unit **207**, and the like depending on the analyzed user operation.

For example, when the analyzed user operation is detection of touch of the lens selection portion (lens selection icon) **105**, information on the lens to be selected according to the touched lens selection portion (lens selection icon) **105** is output to the lens switching control unit **202**.

Further, the operation portion (UI portion) display control unit **208** is notified that the touch of the lens selection portion (lens selection icon) **105** has been detected. The operation portion (UI portion) display control unit **208** executes, for example, processing for displaying the digital zoom ON state identification icon **106** or the digital zoom adjustment section **107** described in the previous embodiment on the display unit **210** in response to this notification of the detection of the touch of the lens selection portion (lens selection icon) **105**.

When the user operation information analyzed by the operation information analysis unit **201** is detection of sliding of the lens selection portion (lens selection icon) **105**, the slide amount is analyzed, and the image processing unit **204** is notified of an analysis result.

The image processing unit **204** determines the digital zoom amount depending on the slide amount input from the operation information analysis unit **201**, and executes image area selection processing from the captured image input from the imaging unit **203** or image enlargement processing for a selection area depending on the determined digital zoom amount.

The digital zoom image generated by the image processing unit **204** is output to and displayed on the image display unit **211** of the display unit **210** via the captured image display control unit **207**.

Further, when the user operation information analyzed by the operation information analysis unit **201** is a shutter operation, the image processing unit **204** is notified that the shutter operation has been performed.

The image processing unit **204** stores the image captured by the imaging unit **203** or a digital zoom processing image thereof in the memory (storage unit) **205** in response to such a shutter operation detection notification.

The lens switching control unit **202** receives information on the lens selection portion (lens selection icon) touched by the user from the operation information analysis unit **201**, and executes processing for switching between lenses to be applied to photographing depending on the input information, as described above.

The imaging unit **203** is configured of an imaging control unit, a plurality of lenses, an image sensor, and the like. A lens to be applied to photographing is changed in response to a lens switching control command input from the lens switching control unit **202**, and the captured image is output to the image processing unit **204**.

The image processing unit **204** performs image processing on the captured image input from the imaging unit **203**, outputs a processed image to the captured image display control unit **207**, and causes the processed image to be displayed on the image display unit **211** of the display unit **210**.

When the slide information of the lens selection portion (lens selection icon) **105** is input from the operation information analysis unit **201**, the image processing unit **204** determines the digital zoom amount depending on the slide amount, and executes the image area selection processing from the captured image input from the imaging unit **203** or the image enlargement processing for the selected area depending on the determined digital zoom amount.

Further, when shutter operation information is input from the operation information analysis unit **201**, the image processing unit **204** stores the image captured by the imaging unit **203** or the digital zoom processing image thereof in the memory (storage unit) **205**.

The captured image display control unit **207** outputs the captured image received from the imaging unit **203** by the image processing unit **204** or the digital zoom processing image generated on the basis of the captured image received from the imaging unit **203** by the image processing unit **204** to the image display unit **211** of the display unit **210** to display the image.

The memory (storage unit) **205** records the captured image received from the imaging unit **203** by the image processing unit **204** or the digital zoom processing image generated on the basis of the captured image received from the imaging unit **203** by the image processing unit **204**.

The memory (storage unit) **205** also stores programs for processing executed in the imaging apparatus **200**, parameters used for various types of processing, and the like.

The operation portion (UI portion) display control unit **208** executes display control for data to be displayed in the operation portion display area **102** described in the previous embodiment.

For example, when the operation information analysis unit **201** notifies that the touch processing with respect to the lens selection portion (lens selection icon) **105** by the user has been detected, the operation portion (UI portion) display control unit **208** executes processing of displaying the digital zoom ON state identification icon **106** or the digital zoom adjustment section **107** described in the previous embodiment on the display unit **210** in response to this touch detection notification.

The operation portion (UI portion) display control unit **208** further executes display control for changing at least one of the output color, shape, and size of the lens selection portion that is slid by the user, depending on the slide amount.

The image display unit **211** of the display unit **210** corresponds to the captured image display area **101** described in the previous embodiment, and displays the captured image of the imaging unit **203** or the digital zoom processing image generated by the image processing unit **204**.

The operation portion (UI portion) display unit **212** of the display unit **210** corresponds to the operation portion display area **102** described in the previous embodiment, and displays an operation portion (UI operation) such as the lens selection portion (lens selection icon) **105**, the digital zoom adjustment section **107**, and the shutter **103**.

The apparatus control unit **215** executes, for example, vibration processing with an intensity corresponding to the slide amount of the lens selection portion (lens selection icon) **105** by the user, music output processing, or the like.

## 7. HARDWARE CONFIGURATION EXAMPLE OF IMAGING APPARATUS

First, the imaging apparatus of the present disclosure can be configured of, for example, a smartphone, a tablet terminal, or a PC having a camera function.

A hardware configuration example of the smartphone, the tablet terminal, or the PC, which is an example of the imaging apparatus of the present disclosure, will be described with reference to FIG. **34**. The hardware illustrated in FIG. **34** is one configuration example of specific hardware of the imaging apparatus of the present disclosure.

A central processing unit (CPU) **301** functions as a control unit or a data processing unit that executes various types of processing according to programs stored in a read only memory (ROM) **302** or a storage unit **308**. For example, the processing according to the sequence described in the above embodiment is executed. A random access memory (RAM) **303** stores a program that is executed by the CPU **301** or data. The CPU **301**, the ROM **302**, and the RAM **303** are connected to each other by a bus **304**.

The CPU **301** is connected to an input and output interface **305** via the bus **304**, and an input unit **306** including a camera, various switches, microphones, sensors, and the like, and an output unit **307** including a display, speakers, and the like are connected to the input and output interface **305**.

The CPU **301** executes various types of processing in response to a command input from the input unit **306** and outputs processing results to the output unit **307**, for example.

A storage unit **308** connected to the input and output interface **305** is configured of, for example, a flash memory, and stores a program executed by the CPU **301** or various types of data. A communication unit **309** includes a proximity communication unit such as NFC, or a communication unit for Wi-Fi communication, Bluetooth (registered trademark) (BT) communication, and data communication via a network such as the Internet or a local area network, and performs communication with an external apparatus.

A drive **310** connected to the input and output interface **305** drives a removable medium **311** such as a magnetic disk, an optical disc, a magneto-optical disc, or a semiconductor memory such as a memory card to execute recording or reading of data.

## 8. CONCLUSION OF CONFIGURATION OF PRESENT DISCLOSURE

Embodiments of the present disclosure have been described in detail above with reference to specific embodiments. However, it is obvious that those skilled in the art can modify or replace the embodiments without departing from the gist of this disclosure. That is, the present invention has been disclosed in the form of examples and should not be construed in a limiting manner. In order to determine the gist of the present disclosure, Claims should be considered.

Further, the technology disclosed in the present specification can have the following configurations.

(1) An imaging apparatus including:

a plurality of lenses available for image capture; and

a display control unit configured to execute control of display data to be output to a display unit,

wherein the display control unit

displays a plurality of lens selection portions corresponding to the plurality of respective lenses on a display unit, and

displays a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

(2) The imaging apparatus according to (1), wherein the plurality of lenses are lenses with different lens magnifications, and

the display control unit

displays a lens magnification corresponding to the lens selection portion in the lens selection portion.

(3) The imaging apparatus according to (1) or (2), wherein the display control unit sequentially changes and displays a lens magnification displayed in the lens selection portion when the user executes slide processing for the lens selection portion.

(4) The imaging apparatus according to any of (1) to (3), wherein the display control unit

displays, in the lens selection portion, a virtual lens magnification enabling capturing of an image after digital zoom processing for the captured image executed depending on the slide amount of the lens selection portion when the user executes slide processing for the lens selection portion.

(5) The imaging apparatus according to any of (1) to (4), wherein the display control unit

outputs an image captured using a lens corresponding to the lens selection portion selected by the user to the display unit in response to the user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

(6) The imaging apparatus according to any of (1) to (5), wherein the display control unit outputs

an image after digital zoom processing for the captured image executed depending on the slide amount of the lens selection portion to the display unit when the user executes slide processing for the lens selection portion.

(7) The imaging apparatus according to any of (1) to (6), further including:

a lens switching control unit configured to set a lens corresponding to the lens selection portion selected by the user as a lens to be applied to photographing processing in response to the user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

(8) The imaging apparatus according to any of (1) to (7), further including:

an image processing unit configured to execute digital zoom processing for the captured image depending on the slide amount of the lens selection portion when the user executes slide processing for the lens selection portion.

(9) The imaging apparatus according to (8), wherein the display control unit displays a digital zoom processing image of the captured image generated by the image processing unit on the display unit.

(10) The imaging apparatus according to any of (1) to (9), wherein the display control unit

executes display control for notifying the user of deterioration of image quality caused by digital zoom processing for the captured image executed depending on the slide amount of the lens selection portion when the user executes slide processing for the lens selection portion.

(11) The imaging apparatus according to (10), wherein the display control unit executes display control for changing at least one of an output color, shape, and size of the lens selection portion that is slid by the user, depending on the slide amount, as display control for notifying the user of the deterioration of the image quality caused by digital zoom processing for the captured image.

(12) The imaging apparatus according to any of (1) to (11),

wherein an apparatus control unit executes vibration processing with intensity according to the slide amount of the lens selection portion, or music output processing with intensity according to the slide amount of the lens selection portion, as display control for notifying the user of deterioration of image quality caused by digital zoom processing for the captured image.

(13) An imaging apparatus control method executed in an imaging apparatus, wherein the imaging apparatus includes

a plurality of lenses available for image capture; and

a display control unit configured to execute control of display data to be output to a display unit,

wherein the display control unit

displays a plurality of lens selection portions corresponding to the plurality of respective lenses on a display unit, and

displays a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

(14) A program for causing imaging apparatus control processing to be executed in an imaging apparatus,

wherein the imaging apparatus includes

a plurality of lenses available for image capture; and

a display control unit configured to execute control of display data to be output to a display unit, and

the program causes the display control unit to execute

processing for displaying a plurality of lens selection portions corresponding to the plurality of respective lenses on a display unit, and

processing for displaying a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit.

Further, a series of processing described in the specification can be executed by hardware, software, or a composite configuration of both. When processing is executed by software, a program in which a processing sequence is recorded can be installed in a memory of a computer built into dedicated hardware and executed, or the program can be installed and executed in a general-purpose computer capable of executing various processing. For example, the program can be recorded on a recording medium in advance. In addition to being installed in a computer from a recording medium, the program can be received via a network such as a local area network (LAN) or the Internet and installed in a recording medium such as an embedded hard disk.

Various processing described in the specification may not only be executed in chronological order according to the description, but may also be executed in parallel or individually according to processing capability of a device that executes the processing or as necessary. Further, in the present specification, the system is a logical collective configuration of a plurality of devices, and the devices of the respective configuration are not limited to being in the same housing.

## INDUSTRIAL APPLICABILITY

As described above, according to the configuration of the embodiment of the present disclosure, the operation portion (UI) that enables efficient execution of the photographing lens selection processing by the user and the digital zoom processing for the captured image using the selected lens is realized.

Specifically, for example, a plurality of lenses with different lens magnifications available for image capturing, and a display control unit configured to execute control of display data to be output to a display unit are included, and the display control unit displays a plurality of lens selection portions corresponding to the plurality of respective lenses on the display unit, displays a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable depending on a slide amount of the lens selection portion selected by a user, in response to a user operation for selecting one of the plurality of lens selection portions displayed on the display unit, and displays a digital zoom image executed depending on the slide amount on the display unit.

With this configuration, the operation portion (UI) that enables efficient execution of the photographing lens selection processing by the user and the digital zoom processing for the captured image using the selected lens is realized.

## REFERENCE SIGNS LIST

**10** Imaging apparatus
**11** Display unit
**12** Camera lens
**101** Captured image display area
**102** Operation portion display area
**103** Shutter
**104** Still image/moving image switching unit
**105** Lens selection portion (lens selection icon)
**106** Digital zoom ON state identification icon
**107** Digital zoom adjustment section
**121** Zoom amount down instruction icon
**122** Zoom amount up instruction icon
**125** Digital zoom ON state notification message
**151** Optical zoomable section identification data
**152** Optical zoom ON state identification icon
**153** Optical zoomable section identification data
**200** Imaging apparatus
**201** Operation information analysis unit
**202** Lens switching control unit
**203** Imaging unit
**204** Image processing unit
**205** Memory (storage unit)
**206** Display control unit
**207** Captured image display control unit
**208** Operation portion (UI portion) display control unit
**210** Display unit
**211** Image display unit
**212** Operation portion (UI portion) display unit
**215** Apparatus control unit
**301** CPU
**302** ROM
**303** RAM
**304** Bus
**305** Input and output interface
**306** Input unit
**307** Output unit
**308** Storage unit
**309** Communication unit
**310** Drive
**311** Removable medium

The invention claimed is:

1. An imaging apparatus comprising:

a plurality of lenses for an image capture process; and

a display control unit configured to:

execute control of display data to be output to a display unit,

display a plurality of lens selection portions corresponding to the plurality of lenses on the display unit; and

display a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable based on a slide amount of a lens selection portion from the plurality of lens selection portions selected by a user, in response to a user operation for selection of the lens selection portion of the plurality of lens selection portions displayed on the display unit.

2. The imaging apparatus according to claim 1, wherein the plurality of lenses are lenses with different lens magnifications, and

the display control unit is further configured to display, in the lens selection portion, a lens magnification corresponding to the lens selection portion.

3. The imaging apparatus according to claim 1, wherein the display control unit is further configured to sequentially change and display a lens magnification displayed in the lens selection portion when the user executes a slide process for the lens selection portion.

4. The imaging apparatus according to claim 1, wherein the display control unit is further configured to display, in the lens selection portion, a virtual lens magnification that enables capturing of an image after a digital zoom process for the captured image executed depending on the slide amount of the lens selection portion when the user executes a slide process for the lens selection portion.

5. The imaging apparatus according to claim 1, wherein the display control unit is further configured to output an image captured by a lens corresponding to the lens selection portion selected by the user to the display unit in response to the user operation for selecting the lens selection portion of the plurality of lens selection portions displayed on the display unit.

6. The imaging apparatus according to claim 1, wherein the display control unit is further configured to output an image after a digital zoom process for the captured image executed depending on the slide amount of the lens selection portion to the display unit when the user executes a slide process for the lens selection portion.

7. The imaging apparatus according to claim 1, further comprising:

a lens switching control unit configured to set a lens corresponding to the lens selection portion selected by the user for a photographing process in response to the user operation for selecting the lens selection portion of the plurality of lens selection portions displayed on the display unit.

8. The imaging apparatus according to claim 1, further comprising:

an image processing unit configured to execute a digital zoom process for the captured image based on the slide amount of the lens selection portion when the user executes a slide process for the lens selection portion.

9. The imaging apparatus according to claim 8, wherein the display control unit is further configured to display a digital zoom processing image of the captured image generated by the image processing unit on the display unit.

10. The imaging apparatus according to claim 1, wherein the display control unit is further configured to execute display control to notify the user of a deterioration of an image quality caused by a digital zoom process for the captured image executed depending on the slide amount of the lens selection portion when the user executes a slide process for the lens selection portion.

11. The imaging apparatus according to claim 10, wherein the display control unit is further configured to execute display control to change at least one of an output color, a shape, or a size of the lens selection portion that is slid by the user, depending on the slide amount, as the display control to notify the user of the deterioration of the image quality caused by the digital zoom process for the captured image.

12. The imaging apparatus according to claim 1, wherein further comprising an apparatus control unit configured to execute one of a vibration process with an intensity associated with the slide amount of the lens selection portion, or a music output process with the intensity associated with the slide amount of the lens selection portion, as display control to notify the user of a deterioration of an image quality caused by a digital zoom process for the captured image.

13. An imaging apparatus control method, comprising:

in an imaging apparatus;

executing, by a display control unit, control of display data to be output to a display unit;

displaying a plurality of lens selection portions corresponding to a plurality of lenses on the display unit; and

displaying a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable based on a slide amount a lens selection portion from the plurality of lens selection portions selected by a user, in response to a user operation for selecting the lens selection portion of the plurality of lens selection portions displayed on the display unit.

14. A non-transitory computer-readable medium having stored thereon, computer-executable instructions which, when executed by a computer, cause the computer to execute operations. the operations comprising:

executing control of display data to be output to a display unit;

displaying a plurality of lens selection portions corresponding to a plurality of lenses on the display unit; and

displaying a digital zoom adjustment section in which a digital zoom amount of a captured image is adjustable based on a slide amount of a lens selection portion from the plurality of lens selection portions selected by a user, in response to a user operation for selecting the lens selection portion of the plurality of lens selection portions displayed on the display unit.

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